1 Approximate laws and the best system

Kepler's laws are false: against a modern planetary theory a fixed ellipse misses Mars by one or two arcminutes over a few decades and the Moon by about a degree within a month (**??**), yet the laws were, and are, good science. An exact law that is slightly wrong is refuted by any datum precise enough to show its error. We ask what SAI does with an approximately true law, and what becomes of Lewis's best system and of explanatory power.[1] M_A also mixes randomised theories, and under SAI's pricing an approximately true law is a law with a calibrated error model (Proposition 1.2); that pricing is not an account of approximate correctness (Section 1.3).

1.1 Setting

Definition 1.1 (Answered data, exception codes, approximate laws)**.** (a) *Answered data.* A datum $(\mathcal{Q}, a)$ is a question (**??**) and its true answer, read as the two-valued datum $(a, 1)$; a data set D has pairwise distinct questions and $n := |D|$ data. T *proves* (*refutes*) the datum if $B \cup T \vdash a$ ($B \cup T \vdash \neg a$), and is *silent* otherwise; $\gamma(\mathcal{Q}) := \log|\mathcal{Q}|$ is the *literal cost.* (b) *Codes.* $\|\Gamma\| := \sum_{\chi\in\Gamma} \ell(\chi)$, with $\ell(\chi) := 1 + |c_S(\chi)|$ for a prefix-free code c_S, for a finite theory or *module* Γ. If $B \cup T \cup D^{\pm}$ is consistent, the *exceptions* of T, the data it does not prove, cost $\mathrm{Exc}^{Q}_D(T) := \sum \gamma(\mathcal{Q})$ (code Q), $\mathrm{Exc}^{F}_D(T) := \sum \ell(a)$ (code F) or $\mathrm{Exc}^{R}_D(T) := -\log P^R_T(D)$, $P^R_T(D)$ the probability of the data when the law's interval is completed uniformly at random (code R, Definition A.1); otherwise $\mathrm{Exc}_D(T) := \infty$. Unmarked, Exc is code Q. The minimisers of $J_\lambda(T; D) := \lambda\|T\| + \mathrm{Exc}_D(T)$, $\lambda > 0$, are the λ-*MAPs* (Definition 3.6 of the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026)); T^{-Z} withdraws T at the times in a finite set Z, at about c_{time} bits each. (c) *Interval laws.* Over $B :=$ RCF with constants q_t for times t in a decidable set I of numerals, a *pointwise interval law* is $\{\underline{f}(t) \le q_t \le \overline{f}(t) : t \in I\}$ for computable $\underline{f} \le \overline{f}$; the *exact law* T_0 has $\underline{f} = \overline{f} = f$, the *honest law* T^{ε} has $f \mp \varepsilon$. *Interval data* answer questions that partition a range $[0, \mathsf{R})$ into cells. (d) *Errors.* $e(t) := f(t) - q(t)$ for a real quantity q, and $\varepsilon^* := \operatorname{ess\,sup}|e|$ at a data time. *Regime G*: intervals of half-width δ around measurements with errors at most $\delta_0 \le \delta$. *Regime R*: the errors $e_j := e(t_j)$ at fresh times t_j are i.i.d., and datum j is the cell of width Δ containing $q(t_j)$ in a uniformly dithered grid on a circle of circumference R; $\gamma = \log(\mathsf{R}/\Delta)$, $p_\Delta := \mathbb{E}\min(|e|/\Delta, 1)$, $s_\Delta := \min(2\varepsilon^*/\Delta, 1)$; e is Euler's number. (e) *Peakedness.* $\mathrm{pk}(e) := \varepsilon^*/\mathbb{E}|e|$ and $\mathrm{pk}_\varepsilon(e) := \varepsilon/\mathbb{E}|e|$; reported values name their fit (minimax or mean).

1.2 Deterministic theories and precision

Over deterministic theories proof-requiring compatibility is monotone in logical strength: $T_0 \vdash T^{\varepsilon}$, data compatible with T^{ε} are compatible with T_0, and with prior weights $2^{-\|T\|}$ the posterior odds of T_0 against T^{ε} stay at least 2, although T_0 is false when $e \not\equiv 0$ (Proposition A.10). In regime G, T_0 is almost surely compatible with the data at every sample size when $\delta \ge \varepsilon^* + \delta_0$ and, under mild conditions on the measurement error, eventually refuted when $\delta < \varepsilon^* + \delta_0$ (Proposition A.12). In regime R the exact law survives n data with probability $(1 - p_\Delta)^n$ and is almost surely refuted eventually iff $\mathbb{P}[e \ne 0] > 0$, however coarse the cells; with a sinusoidal error of amplitude $1'$ it is refuted by positions rounded to a dithered grid of 1° after about 94 data (Proposition A.13).

[1] The questions of approximately true laws and of the domain of applicability of a model are discussed in Hänni's working notes (Hänni, 2026).

1.3 Randomised theories and laws with error models

A theory machine reading a random tape answers each question at random within the law, so $L_A(D) \le \|T\| + \mathrm{Exc}_D^R(T) + c$ for a pointwise interval law T on interval data consistent with it (Proposition A.4). SAI is the best two-part code over theory machines, $L_A(D) = \min_G[K(G) - \log \nu_G(D)] \pm c$ (Proposition A.5); *SAI's pricing* of a form is the weight $2^{-(K(G)-\log \nu_G(D))}$ of a theory machine G in M_A's mixture. Between the honest law T^{ε^*} and the exact law with exceptions T_0^{-Z}, for $\Delta \ge 2\varepsilon^*$, spread errors favour the first and peaked ones the second, with a threshold on $\mathrm{pk}(e)$ set by the prices of an exception and of silence (Propositions A.17 and A.18); band-confined completions move this comparison inside the band. For $\varepsilon \ge \varepsilon^*$ and a computable distribution $\mathcal{G}$ on $[0,1]$, the *band-confined completion* $\mathrm{BC}_\mathcal{G}$ states T^ε and draws each value $f(t_i) - \varepsilon + 2\varepsilon\hat{w}_i$, $\hat{w}_i \sim \mathcal{G}$, from the tape; at fresh times its law is $\prod_j \mathcal{G}(C'_j)$, C'_j the cell of datum j with the band mapped affinely onto $[0,1]$. Uniform $\mathcal{G}$ gives the *uniform band*, and $\mathcal{G} = (1-\rho)\delta_{1/2} + \rho\,\mathrm{Unif}[0,1]$ gives $\mathrm{HY}(\rho)$: exact except in a share ρ of cases and then within ε.

Proposition 1.2 (Laws with error models)**.** *In regime R let $\varepsilon \ge \varepsilon^*$, and let $\mathcal{F}$ be the distribution function of $w_j := (\varepsilon - e_j)/(2\varepsilon)$, the position of the true value in the band.*

(a) If $\Delta \ge 2\varepsilon$, the rate of $\mathrm{BC}_\mathcal{G}$ is $(2\varepsilon/\Delta)\, c_\mathcal{G}$, with

$$c_\mathcal{G} := \mathbb{E}_w \int_0^1 \left(\mathbf{1}[v < w](-\log(1 - \mathcal{G}(v))) + \mathbf{1}[v > w](-\log \mathcal{G}(v))\right) dv.$$

So band-confined completions are ranked in the same way at every such Δ.

(b) $c_\mathcal{G} \ge c_\mathcal{F} := \int_0^1 H_2(\mathcal{F}(v))\, dv$, H_2 the binary entropy, with equality iff $\mathcal{G} = \mathcal{F}$ almost everywhere on $[0,1]$. In particular the uniform band is the best band-confined completion iff e is uniform on $[-\varepsilon, \varepsilon]$.

(c) For every theory machine G', G_U included, almost surely $\liminf_n \frac{1}{n}(-\log \nu_{G'}(D_n)) \ge r_\mathcal{F}(\Delta) := \mathbb{E}[-\log \mathcal{F}(C'_1)]$, the conditional entropy rate of the answers given the questions, while $-\frac{1}{n}\log \nu_{\mathrm{BC}_\mathcal{F}}(D_n) \to r_\mathcal{F}(\Delta)$. If $\mathcal{F}$ is computable, then $L_A(D_n)/n \to r_\mathcal{F}(\Delta)$ almost surely. For $\Delta \ge 2\varepsilon$, $r_\mathcal{F}(\Delta) = (2\varepsilon/\Delta)\, c_\mathcal{F}$.

(d) $c_{\mathrm{HY}(\rho)}$ is convex in $\rho \in (0,1]$, and some $\mathrm{HY}(\rho)$ with $\rho < 1$ beats the uniform band iff $\mathbb{E}\ln(1 + |e|/\varepsilon) < 2\ln 2 - 1$. This requires $\mathrm{pk}_\varepsilon(e) > \ln 2/(2\ln 2 - 1) \approx 1.79$, and holds whenever $\mathrm{pk}_\varepsilon(e) > \mathrm{e}/(4 - \mathrm{e}) \approx 2.12$.

Proof idea. A boundary falls in the band with probability $2\varepsilon/\Delta$, and the datum costs $-\log$ of the $\mathcal{G}$-mass on the true side; (b) is Gibbs' inequality, (c) the Borel–Cantelli lemma (Appendix A.4).

So under SAI's pricing the preferred form is a law with a calibrated error model, neither a silent partial theory nor an exact law with listed exceptions. That a model includes its noise distribution is standard in MDL (cf. Rissanen, 1978; Grünwald, 2007); what is specific is that for a proof-requiring inducer the honest partial theory is one completion among many. Selection among finitely many forms is almost sure, and for the randomised theories that M_A mixes humility is calibration (Theorem A.22, Corollary A.23, and Proposition A.24).

What the pricing does not see. Proposition 1.2 describes the credence- and compression-best forecast, not approximate correctness. For questions about one quantity the loss is the log score of the cell that the answers determine: a question family fixes a resolution, while the cost of a miss is set by each hypothesis's own error model, not by a purpose (Proposition A.42). SAI's pricing ranks rival laws by a dispersion of the error for coarse cells and, for fine ones, by the entropy of

the error rather than its size (Theorem A.45); it can rank a law above one that is closer to the truth at every time; and other proper scores rank rival laws differently, under misspecification and even without it (Examples A.47 and A.48). A tolerance on a domain is a sentence, in which SAI holds a credence; a graded loss is not (Proposition A.49).

Kepler's residuals. For seven bodies and six windows each (three for the Moon) we fitted one Kepler ellipse four ways and kept the fit with the least largest longitude error (the minimax fit) and the one with the least mean error (the mean fit), with errors against the plan94 and moon98 ephemerides (Table 1). A computation at eleven cell widths (code in the accompanying repository) finds that among the Kepler-based theory machines compared a law with a 40-bin error histogram, fitted and scored on disjoint halves of the times, has the least rate in all 429 cells, and that in the coarse regime the two-point form beats the uniform band for 34 of 39 body–window pairs, exactly where Proposition 1.2(d) says. Over the four fits, maximum, RMS and mean absolute error disagree on the best fit, the calibrated pricing agrees with none, and its ranking changes with cell width for 11–16 of 234 pairs, by up to 6.8% of the rate (Appendix A.9).

1.4 Sound reasoning with approximate models

True honest models of one quantity are jointly consistent over a true background and prove only truths about it; when the background decides the comparisons (as RCF does for semialgebraic models), exact approximations that differ anywhere on their common domain are jointly inconsistent and so prove everything (Proposition A.28). Honest dynamical laws have derivation horizons, up to which exact approximations are guaranteed correct (Propositions A.25 and A.27); after 100 steps with Lipschitz constant 1 the propagated tolerance is 100ε, against a mean deviation of about 7.96ε for independent random-sign errors (a computation, Remark A.26). The device does not say how to choose a model, a domain or a tolerance (**??**).

1.5 The best system as a steeper MAP

Lewis's laws are the theorems of the system that best balances simplicity against strength (cf. Lewis, 1983, 1994). By analogy (best systems are a metaphysics of laws, SAI is inference from finite data) we take strength to be fit to data in a fixed base language, the balance to be λ, and laws with private literal facts as candidates. The points $(\|T\|, \mathrm{Exc}_D(T))$ have a convex hull with vertices v_i in order of length and edge slopes $-\sigma_i$, $\infty =: \sigma_0 > \sigma_1 > \cdots > 0$; for $\lambda \in (\sigma_{i+1}, \sigma_i)$ the λ-MAPs are the theories at v_i (Lemma 3.8 of the companion paper on steeper priors). A point $(\alpha_{\mathrm{r}}, L_{\mathrm{r}})$ of some theory is a *zero-window* (σ, τ)*-kink* (Definition 5.2 there) if theories shorter by $x > 0$ have fit at least $L_{\mathrm{r}} + \sigma x$ and theories longer by $x > 0$ fit at least $L_{\mathrm{r}} - \tau x$.

Proposition 1.3 (Robustness to the balance is a kink)**.** *For a family of candidate theories satisfying the hypotheses of the hull lemma (Proposition A.8) and $0 \le \tau < \sigma \le \infty$, the following are equivalent: (a) the set of λ-MAPs is the same for every $\lambda \in (\tau, \sigma)$; (b) some hull vertex v_i has $\sigma_{i+1} \le \tau$ and $\sigma \le \sigma_i$; (c) the point family has a zero-window (σ, τ)-kink at a point that is a hull vertex.*

Proof idea. The λ-MAP set is fixed between consecutive edge slopes and changes at each; (b)⇔(c) is Corollary 5.4(c) of the companion paper on steeper priors (Appendix A.6).

So the best system is the same for every balance in an interval exactly when the data's hull has a kink there. This makes precise robustness to the balance of simplicity against strength, with

the measures of simplicity and of fit held fixed, one aspect of what Lewis calls a kind nature (cf. Lewis, 1994); robustness to the measure of simplicity itself fails among the compatible Mars forms in the noise-free regime (**??**). On a solar-system toy a computation finds one best system for all $\lambda \in (7.03, 135.37)$ (Appendix A.6); Lewis's worry about a predicate true of exactly the actual things is blocked up to the complexity of a language chosen after the data (Proposition A.31).

The *explanatory power* $\mathrm{EP}_D(\mathcal{A}; T)$ of a module $\mathcal{A} \subseteq T$ is the bits of data it explains per bit of its statement, in the context T (Definition A.29). Every module of a λ-MAP has EP $\geq \lambda$, no addition to it has EP $> \lambda$, and for $\lambda > 1$ a fact used once never enters (Propositions A.30 and A.32). So the model–randomness split of the companion paper on steeper priors separates facts reused across many data from facts used once, not laws from initial conditions; what singles out a law is that its EP grows with the number of systems it serves. Kepler's third law, which shortens the stated elements, is a law about initial conditions (Proposition A.33; for its worth to Kepler, **??**), and in the question format elements never enter a λ-MAP (Proposition A.34).

1.6 Explanation and correction

$\|\mathcal{A}\| \, \mathrm{EP}_D(\mathcal{A}; T)$ is the log-likelihood ratio of T against $T \setminus \mathcal{A}$ for $P_T(D) := 2^{-\mathrm{Exc}_D(T)}$. The support a hypothesis gets from joint data $D_1 \cup D_2$ splits into the supports from the parts plus a term U, the change in the mutual information of D_1 and D_2 that the hypothesis brings about, of the form of Myrvold's measure of unification (Myrvold, 2003): $U = 0$ for laws that are products over questions with stated parameters; with a k-bit parameter integrated out, U is k within one system when each part of the data identifies the parameter, 0 across systems with independent parameters, and the bits of a shared constant across systems that share one and each identify it (Proposition A.35). Copernicus against Ptolemy is unification without better prediction (Proposition A.36).

Newton's theory does not entail Kepler's laws; with the planets' masses and initial conditions it contradicts their exact form (cf. Duhem, 1954) and derives corrected statements. In an exact rational toy of perturbed circular motion the successor proves the honest old law with uniform tolerance ε on a window iff ε is at least the largest distance between the two theories' trajectories on it, and refutes the exact old law whenever the perturbation changes the first step (Proposition A.37): the old law's domain of applicability is derived, not stipulated.

1.7 Domains of applicability and retention

Domains are learned as tolerances. A domain restriction leaves a system's data literal (Proposition A.39). Under SAI's pricing an honest band of half-width ε^* costs at most $\log(2\varepsilon^*/\Delta) + \log \mathrm{e}$ bits per datum when $\Delta \leq 2\varepsilon^*$, below the literal $\log(\mathsf{R}/\Delta)$ iff $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$, and at most $(2\varepsilon^*/\Delta) \log \mathrm{e} < \log(\mathsf{R}/\Delta)$ when $2\varepsilon^* \leq \Delta < \mathsf{R}/\mathrm{e}$; so the Moon ($\varepsilon^* = 2.6°$ over a year, minimax fit) is kept in Kepler's law with its own tolerance at every cell width computed, while literal pricing at $10'$ and $60'$ excludes it by a cheap domain clause. We conclude that under SAI's pricing a domain of applicability shows up as a tolerance and an error distribution that depend on the system and the window, not as a sharp boundary; a superseded theory is retained where its derivation savings exceed its loss of fit (Proposition A.40).

A Proofs for approximate laws and the best system

This appendix states in full the results of Section 1 that the main text states in words, and proves every result of that section. The notation is that of Definitions 1.1 and A.1. We measure a finite

theory by $\|T\|$, without the stop bit of a prefix-free code of theories; using the prefix-free length $\|T\|+1$ instead changes every statement below by an additive constant at most, and shifts J_λ by the same amount λ on every theory, so that it leaves the λ-MAPs unchanged. Computed numbers come from seeded scripts in the accompanying repository; they are never part of a statement.

A.1 The setting in full

Definition A.1 (The setting in full, continuing Definition 1.1)**.**

(a) *Answers and objectives.* $\operatorname{dom} D$ lists the questions of a data set D in a canonical order, and y_D their answers. $A_T(\mathcal{Q}) := \{a' \in \mathcal{Q} : B \cup T \nvdash \neg a'\}$ is the set of answers that T allows. J^Q_λ, J^F_λ and J^R_λ are J_λ with Exc^Q, Exc^F and Exc^R.

(b) *Interval laws.* $B = \mathrm{RCF}$ is decidable and complete (standard; Tarski, 1951), and R is rational. The total computable maps $\underline{f} \le \overline{f} : I \to \mathbb{Q} \cap [0, \mathsf{R}]$ of a pointwise interval law are named by a codeword of a fixed decidable prefix-free presentation language, and $\|T\|$ is the codeword's length. The honest law T^ε has $\underline{f} = f - \varepsilon$ and $\overline{f} = f + \varepsilon$, clipped to $[0, \mathsf{R}]$. The questions of interval data partition $[0, \mathsf{R})$ into finitely many cells with rational endpoints. With C_t the intersection of the cells that D assigns to the time t,

$$P^R_T(D) := \prod_t \frac{|C_t \cap [\underline{f}(t), \overline{f}(t)]|}{\overline{f}(t) - \underline{f}(t)}$$

($|\cdot|$ the length; the factor is $\mathbf{1}[\underline{f}(t) \in C_t]$ if $\underline{f}(t) = \overline{f}(t)$), the probability that values drawn uniformly and independently from the law's intervals lie in the cells of the data.

(c) *Regime G.* Datum j is "$q(t_j) \in [v_j - \delta, v_j + \delta]$", with $v_j = q(t_j) + \zeta_j$ rational and $|\zeta_j| \le \delta_0 \le \delta$.

(d) *Regime R.* q lives on a circle of circumference R. Datum j is about a *fresh* time t_j (the t_j form an injective computable sequence of numerals); the errors $e_j := e(t_j)$ are i.i.d.; and datum j is the cell of width Δ (R/Δ an integer) that contains $q(t_j)$ in a grid shifted by a dither u_j, uniform and independent of everything else.

Technical points. (i) *Rational cells.* In regime R the dither u_j is uniform on a grid $\{k/N_\mathrm{d} : 0 \le k < N_\mathrm{d}\}$, so that cell endpoints are rational. Every event used below has the form "u_j lies in an interval of length ℓ'", whose grid probability is within $1/N_\mathrm{d}$ of ℓ'; the formulas are the limit $N_\mathrm{d} \to \infty$ (equivalently, endpoints may be real algebraic numbers). A cell that crosses the zero point of the circle is a disjunction of two interval sentences. (ii) *Fresh times.* Data are sentences, so they mention numerals, a countable set; no atomless law lives on it. Times drawn i.i.d. from a law on numerals repeat: the number of distinct times among n draws is $o(n)$ almost surely (take a finite set of mass at least $1 - \epsilon$ and apply the strong law of large numbers). A repeat changes the cost of every form except an exact law: a patched law withdraws the law at a whole time, so every later datum at a refuted time is an exception, and the cells of an honest law at one time intersect, which narrows its band. A computation with times drawn uniformly from the 4383 days of a twelve-year window (sinusoidal error, $\varepsilon^* = 1'$, $\Delta = 60'$, $\lambda = 1$, $c_\mathrm{time} = \log 4383 + 2$) gives a patched law's cost per datum of 0.40, 1.96 and 3.23 bits at $n = 10^3$, 10^5 and 2×10^5, against the rate 0.240 at fresh times, and a code-R cost of the honest law falling from 0.037 to 0.022, against 0.030. This is why regime R takes fresh times: an injective computable sequence of numerals, with i.i.d. errors and dithers. Fresh times have names that grow, and Proposition A.17 says what that costs. (iii) *Presentations.* No decidable set of programs covers all total computable maps, so

the class of pointwise interval laws depends on the presentation language (for example, primitive recursive definitions, or programs paired with an explicit clock); every statement holds for each fixed choice. (iv) *A model of the data.* In regime R the error at a fresh time is a draw from a fixed law; for the Kepler residuals this law is the law of $e(t)$ for t uniform in the window. Answered data with a question $\mathcal{Q}$ are read through **??**: for consistent $B \cup T$, T proves $(\mathcal{Q}, a)$ iff $A_T(\mathcal{Q}) = \{a\}$, since $T \vdash a$ refutes every other answer by incompatibility, and refuting every other answer proves a because B proves $\bigvee \mathcal{Q}$.

Lemma A.2 (Decidability)**.** *Let T be a pointwise interval law and C a cell, a finite union of intervals with rational endpoints. Then T proves the datum "$q_t \in C$" iff $[\underline{f}(t), \overline{f}(t)] \subseteq C$, and refutes it iff $[\underline{f}(t), \overline{f}(t)] \cap C = \emptyset$; and $B \cup T \cup D^{\pm}$ is consistent iff, for every time t of D, $[\underline{f}(t), \overline{f}(t)]$ meets the intersection C_t of the cells that D assigns to t. All three are decidable from the presentation of T.*

Proof. Every choice of reals $q_t \in [\underline{f}(t), \overline{f}(t)]$ gives a model of $B \cup T$ in $\mathbb{R}$, and if the q_t also lie in the assigned cells, a model of $B \cup T \cup D^{\pm}$. This gives the "only if" of the first two claims (choose q_t outside the cell, respectively inside it) and the "if" of the third. Inclusion and disjointness of finite unions of rational intervals are Boolean combinations of inequalities between rational numerals, which RCF proves or refutes; this gives the "if" of the first two claims. If for some t the interval misses C_t, the finitely many inequalities involved are jointly refutable in RCF, which gives the "only if" of the third. The values $\underline{f}(t), \overline{f}(t)$ are computable from the presentation, so all three conditions are decidable. □

Proposition A.3 (Laws plus listed exceptions are deterministic hypotheses)**.**

(a) For every finite T with $B \cup T \cup D^{\pm}$ consistent, $L_A(D) - c \le K_A(D) \le \|T\| + \mathrm{Exc}_D^F(T) + c$.

(b) For an r.e. T and $E \subseteq D$ with $B \cup T \cup E^{\pm}$ consistent and $B \cup T \cup E^{\pm} \vdash \psi_D$, $L_A(D) \le K_{\equiv}(T) + K(E) + c$.

Proof. (a) The finite set $T' := T \cup \{a : (\mathcal{Q}, a)$ a datum of D that T does not prove$\}$ is contained in the consistent set $T \cup D^{\pm}$, so $B \cup T'$ is consistent, and T' proves every datum. Concatenating the codewords of the sentences of T, each after a continuation bit, the codewords $c_S(a)$ of the exceptions, each after a continuation bit, and a stop bit gives a prefix-free code of T' of length $\|T\| + \mathrm{Exc}_D^F(T) + 1$, and a fixed program turns it into an index of T'. So $K_A(D) \le \|T\| + \mathrm{Exc}_D^F(T) + c$. The instance form of dominance (**??**) gives $M_A(D) \ge 2^{-K(e)-c}$ for every index e of a theory consistent with B that proves ψ_D, hence $L_A(D) \le K_A(D) + c$. (b) is **??**(b). □

So code F pays for *which* questions are exceptions, since the answer sentence names its question; a deterministic theory cannot avoid this, because it does not see the questions.

A.2 Randomised theories and the best two-part code

Proposition A.4 (Randomised theories pay question-relative costs)**.** *Let T be a pointwise interval law and D a set of interval data consistent with T.*

(a) $L_A(D) \le K(T) + \mathrm{Exc}_D^R(T) + c \le \|T\| + \mathrm{Exc}_D^R(T) + c$.

(b) If D comes from a computable question family with one question $\mathcal{Q}_t$ per time, and $k_t(T) := |A_T(\mathcal{Q}_t)| \le |\mathcal{Q}_t|$, then $L_A(D) \le K(T, (\mathcal{Q}_t)_t) + \sum \log k_t(T) + c$, the sum over the data that T does not prove; in particular $L_A(D) \le K(T, (\mathcal{Q}_t)_t) + \mathrm{Exc}_D^Q(T) + c$.

(c) *Code R charges an exception of a patched law, at a time where $\underline{f} = 0$ and $\overline{f} = \mathsf{R}$, exactly $\log(\mathsf{R}/|C_t|)$, which is the code-Q cost for equal cells; it charges the honest law $-\log(|C_t \cap [f(t)-\varepsilon, f(t)+\varepsilon]|/2\varepsilon)$, which is 0 when the band lies inside the cell.*

Proof. (a) *The machine.* Enumerate I as $t_1, t_2, \ldots$ and fix a computable injection $(i,k) \mapsto \langle i,k\rangle$ into tape positions, increasing in k. On a tape ω put $U_i := \sum_{k\geq 1} \omega_{\langle i,k\rangle} 2^{-k}$; under $\mathbb{U}$ the U_i are independent and uniform on $[0,1]$. Let G_T be the theory machine that emits the sentences of T at the empty prefix and, once the prefix read contains the positions $\langle i,1\rangle, \ldots, \langle i,k\rangle$, whose bits form the binary numeral of an integer m, also emits

$$\underline{f}(t_i) + (\overline{f} - \underline{f})(t_i)\, m 2^{-k} \;\leq\; q_{t_i} \;\leq\; \underline{f}(t_i) + (\overline{f} - \underline{f})(t_i)\,(m+1)2^{-k}.$$

This set of (prefix, sentence) pairs is c.e., and an index of it is computable from the presentation of T, so $K(G_T) \leq K(T) + c$.

The random theory is always consistent. Put $q_{t_i} := v_i := \underline{f}(t_i) + (\overline{f} - \underline{f})(t_i) U_i$. Each emitted closed interval contains v_i, so these values satisfy every emitted sentence, and $\mathbb{R}$ with them is a model.

It proves a datum "$q_{t_i} \in C$" whenever v_i lies in the interior of C. The emitted intervals contain v_i and have lengths $(\overline{f} - \underline{f})(t_i) 2^{-k} \to 0$, so one of them lies in the interior of C, and RCF proves the inclusion. (If $\underline{f}(t_i) = \overline{f}(t_i)$, T alone decides the datum.) The times of D are distinct indices i, the U_i are independent, and endpoints have probability 0, so

$$\nu_{G_T}(D) \;\geq\; \mathbb{U}\{\omega : v_t \in \operatorname{int} C_t \text{ for every time } t \text{ of } D\} \;=\; P^R_T(D).$$

Dominance (**??**), $M_A \geq 2^{-K(\nu)}\nu$ for every theory-machine law ν, gives $M_A(D) \geq 2^{-K(G_T)-c}\nu_{G_T}(D)$, which is (a); and $K(T) \leq \|T\| + c$.

(b) Let the machine use U_i to pick the $\lfloor k_{t_i}(T) U_i \rfloor$-th element a of $A_T(\mathcal{Q}_{t_i})$, which is computable by Lemma A.2, and emit a as soon as enough bits have been read (almost surely after finitely many). The cell a meets $[\underline{f}(t_i), \overline{f}(t_i)]$, so the random theory has a model in $\mathbb{R}$; it proves the true answer with probability $1/k_{t_i}(T)$, independently across times, and $k_t(T) = 1$ when T proves the datum. Since D is consistent with T, T refutes no datum. The machine's index is computable from T and the question family, and dominance gives the bound; the last claim follows from $\log k_t(T) \leq \log|\mathcal{Q}_t| = \gamma(\mathcal{Q}_t)$. (c) is the definition of P^R_T. □

Proposition A.5 (SAI is the best two-part code over theory machines)**.**

(a) *For every data set D, $L_A(D) = \min_G[K(G) - \log \nu_G(D)] \pm c$, the minimum over theory machines G; it is attained, up to c, by G_U.*

(b) *Restricting G gives upper bounds: $L_A(D) \leq \min_T J^R_1(T;D) + c$ over pointwise interval laws, $L_A(D) \leq \min_T J^F_1(T;D) + c$ over finite theories, and $L_A(D) \leq K_A(D) + c$.*

(c) *Neither K_A nor $\min_T J^Q_1(T;\cdot)$ over finite theories is within an additive constant of L_A in general; nor, therefore, is $\min_T J^F_1(T;\cdot)$ when $\gamma \leq \ell$.*

(d) *Over all theory machines steeper selection collapses. Take the point family of programs $p \in \operatorname{dom} V$ for theory machines, with length $|p|$ and fit $-\log \nu_{G_{V(p)}}(D)$, and let ℓ_U be the length of a program for G_U. For $\lambda > 1$, every λ-MAP p has $|p| \leq \ell_U + \ell_U/(\lambda - 1)$.*

Proof. (a) Dominance gives $L_A(D) \leq K(G) - \log \nu_G(D) + c$ for every theory machine G, and $M_A = \nu_{G_U}$ gives $K(G_U) - \log \nu_{G_U}(D) = L_A(D) + O(1)$. (b) is Propositions A.3 and A.4 with $K_{\equiv}$ and K of a finite theory at most its length plus a constant.

(c) That $K_A - L_A$ is unbounded is the absence of a coding theorem (**??**). For $\min_T J_1^Q(T;\cdot)$, take the language $\{R\} \cup \{c_j : j \in \mathbb{N}\}$ with $B = \emptyset$ and the yes/no data $D_n := \{(R(c_j), 1) : j < n\} \cup \{(\exists x\, \neg R(x), 1)\}$, so that every datum has literal cost $\gamma = 1$. By **??**, applied to the r.e. theory $\{R(c_j) : j \in \mathbb{N}\} \cup \{\exists x\, \neg R(x)\}$, $K_A(D_n) \le c$, hence $L_A(D_n) \le K_A(D_n) + c \le c'$. Let T be finite with $B \cup T \cup D_n^{\pm}$ consistent. If c_j does not occur in T and $T \vdash R(c_j)$, then $T \vdash \forall x\, R(x)$ by the theorem on constants, which contradicts consistency with $\exists x\, \neg R(x)$. So T proves at most the data $R(c_j)$ with c_j occurring in T, together with $\exists x\, \neg R(x)$. In a syntax code in which each symbol occurrence costs at least one bit, each constant occurring in T costs at least one bit of $\|T\|$, and each unproved datum one bit of Exc^Q; so $J_1^Q(T; D_n) \ge n$. Finally $\mathrm{Exc}^F \ge \mathrm{Exc}^Q$ when $\gamma \le \ell$, since then each exception costs at least as much in code F.

(d) Proposition 5.7(ii) of the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026) states: if a point family contains a member π_U, of length ℓ_U, with fit $L(\pi_U) \le L(\pi) + \ell(\pi) + \kappa_0$ for every member π, then for every $\lambda > 1$ every λ-MAP has length at most $\ell_U + (\ell_U + \kappa_0)/(\lambda - 1)$. The mixture identity of **??** gives $M_A(D) \ge \sum_{p \in \mathrm{dom}\, V} 2^{-|p|} \nu_{G_{V(p)}}(D) \ge 2^{-|p|} \nu_{G_{V(p)}}(D)$ for each p, so a program p_U for G_U has fit $-\log \nu_{G_U}(D) = L_A(D) \le -\log \nu_{G_{V(p)}}(D) + |p|$ for every p. This is the hypothesis with $\kappa_0 = 0$. □

So over all theory machines a steeper penalty collapses onto the universal machine, and the best-system analyses of Section 1.5 use laws with private literal facts instead.

Remark A.6 (Pointwise interval laws: the separation is open). For every set D of interval data, both $L_A(D)$ and $\min_T J_1^R(T; D)$ over pointwise interval laws are at least $K(y_D \mid \mathrm{dom}\, D) - c$. For L_A: the answer conjunctions of the questions in $\mathrm{dom}\, D$ form a finite set S of pairwise B-incompatible sentences, computable from $\mathrm{dom}\, D$, and $M_A(D) = M_A(\{(\psi_D, 1)\})$ (**??**); so the coding lower bound of **??** gives $L_A(D) \ge K(\psi_D \mid S) - c \ge K(y_D \mid \mathrm{dom}\, D) - c'$. For J_1^R: Proposition A.9(b). So neither bound separates the two. The natural candidate for a separation, data in which pairs of times share a random cell, has $L_A \approx n\gamma/2$; but it does not separate them either, because a presentation may store values: the program that stores the $n/2$ shared cells and returns each for both times of its pair has length $n\gamma/2 + O(\log n)$ and $\mathrm{Exc}^R = 0$, so $\min_T J_1^R \le n\gamma/2 + O(\log n)$ as well.

Question A.7. Fix a presentation language for pointwise interval laws. Is $\min_T J_1^R(T; D) - L_A(D)$ bounded over all sets D of interval data? (For K_A and for $\min_T J_1^Q$ over finite theories the answer is no, by Proposition A.5(c).)

Proposition A.8 (Which hypotheses of the companion paper on steeper priors hold here)**.**

(a) Pointwise interval laws in code R *(a fixed presentation language; one question family with finitely many cells). The codewords form a decidable prefix-free set; P_T^R is computable uniformly in T, is the law ν_{G_T} of a theory machine (a shared hypothesis in the sense of Definition 3.4 of the companion paper on steeper priors) and, on data at distinct times, a product hypothesis (Definition 3.3 there); Kraft's inequality holds, lengths are integers, and finitely many laws have each length. So Lemmas A.1, 3.8, 3.9 and 5.3, Corollaries 5.4 and 5.5 and Theorem 5.6 of the companion paper on steeper priors apply, with $J_\lambda = J_\lambda^R$.*

(b) Finite theories over a decidable consistent B, in code Q, *with $P_T(D) := 2^{-\mathrm{Exc}_D^Q(T)} \mathbf{1}[B \cup T \cup D^{\pm}$ consistent$]$. P_T satisfies the semimeasure inequalities of Definition 3.4 of the companion paper on steeper priors, and it is a product over questions when consistency can be checked question by question (pointwise laws at distinct times). It is not lower semicomputable in general (for $B = \emptyset$ and a language with a binary relation symbol), so this class*

is not a hypothesis class of the companion paper. Its hull lemma (Lemma 3.8), kink lemma (Lemma 5.3), Corollary 5.4, Theorem 5.6 and Lemma A.1(i) use only the point family $(\|T\|, \mathrm{Exc}^Q_D(T))_T$ and apply. Its coding bound (Lemma 3.9) is replaced by Proposition A.9, with which Corollary 5.5 holds with offset $b+\kappa+n$ in general, and $b+\kappa$ when provability of the answers from the theories is decidable (κ the constant of Proposition A.9).

Proof. Which hypotheses are used. Each listed property is checked in the statement: for (a), $P^R_T = \nu_{G_T}$ on interval data by the proof of Proposition A.4(a) (the random theory proves "$q_t \in C$" iff $v_t \in \operatorname{int} C$, up to null events), and the product form holds because the U_i are independent. That the hull, kink and converse results need only the point family can be read off their statements, which concern countable point families, and their proofs, which use only lengths, fits, and the finiteness of the set of programs of each length.

The semimeasure inequalities in (b). $P_T(\emptyset) = \mathbf{1}[B \cup T \text{ consistent}] \le 1$. Let $\mathcal{Q} \notin \operatorname{dom} D$ and $P_T(D) > 0$. If T proves an answer a_0, every other answer is refuted by T and contributes 0, while a_0 adds no cost and keeps consistency; so the sum is $P_T(D)$. If T proves no answer, each of the at most $|\mathcal{Q}|$ answers consistent with $B \cup T \cup D^{\pm}$ contributes at most $2^{-\gamma(\mathcal{Q})} P_T(D) = P_T(D)/|\mathcal{Q}|$. If $P_T(D) = 0$, every term is 0.

Not lower semicomputable. Take $B = \emptyset$ and a language with a binary relation symbol. If $T \mapsto P_T(\emptyset)$ were lower semicomputable uniformly in T, the set of finite theories with $P_T(\emptyset) > \frac{1}{2}$, that is the consistent ones, would be r.e. The inconsistent ones are r.e. as well, so consistency of finite theories would be decidable, contrary to Church's theorem (standard; Church, 1936). □

Proposition A.9 (Coding bound for laws with literal exceptions)**.** *Let $n := |D|$.*

(a) For every finite T with $B \cup T \cup D^{\pm}$ consistent, $\|T\| + n + \mathrm{Exc}^Q_D(T) \ge K(y_D \mid \operatorname{dom} D) - c$.

(b) Suppose that provability of answers from T is decidable uniformly in T, for T ranging over a class of presented theories with $K(T) \le \|T\| + c$ (finite theories over a decidable complete B, or pointwise interval laws). Then $\|T\| + \mathrm{Exc}^Q_D(T) \ge K(y_D \mid \operatorname{dom} D) - c$ and $\|T\| + \mathrm{Exc}^R_D(T) \ge K(y_D \mid \operatorname{dom} D) - c$ for every member T consistent with D.

Proof. (a) For a question $\mathcal{Q}$ put

$$w_T(a \mid \mathcal{Q}) := \tfrac{1}{2}\mathbf{1}[a \text{ is the first answer of } \mathcal{Q} \text{ found provable from } B \cup T] + \tfrac{1}{2}|\mathcal{Q}|^{-1},$$

where "first" refers to a fixed dovetailed proof search. Then $\sum_a w_T(a \mid \mathcal{Q}) \le 1$, and w_T is lower semicomputable uniformly in $(T, \mathcal{Q})$, because the first answer found never changes. So $P'_T(y) := \prod_j w_T(y_j \mid \mathcal{Q}_j)$ is a semimeasure on the answer lists for $\operatorname{dom} D$, lower semicomputable uniformly in $(T, \operatorname{dom} D)$. If $B \cup T$ is consistent and proves y_j, it proves no other answer of $\mathcal{Q}_j$, so y_j is the first found and $w_T(y_j \mid \mathcal{Q}_j) \ge \frac{1}{2}$; otherwise $w_T(y_j \mid \mathcal{Q}_j) \ge \frac{1}{2}|\mathcal{Q}_j|^{-1}$. Hence $-\log P'_T(y_D) \le n + \mathrm{Exc}^Q_D(T)$. The conditional coding theorem ($K(y \mid z) \le -\log P(y \mid z) + c_P$ for a semimeasure $P(\cdot \mid z)$ lower semicomputable uniformly in z; standard, see Li and Vitányi, 2008) gives $K(y_D \mid T, \operatorname{dom} D) \le n + \mathrm{Exc}^Q_D(T) + c$, and $K(y_D \mid \operatorname{dom} D) \le K(T) + K(y_D \mid T, \operatorname{dom} D) + c \le \|T\| + n + \mathrm{Exc}^Q_D(T) + c'$.

(b) With decidable provability put $w_T(a \mid \mathcal{Q}) := 1$ if a is the first answer, in a fixed order of $\mathcal{Q}$, that T proves; $:= 0$ for the other answers that T proves; and $:= |\mathcal{Q}|^{-1}$ for every answer if T proves none. This sums to at most 1 for every T, consistent or not, and it is computable. For T consistent with D the cost bound of (a) holds without the flag bit, and the same coding argument gives the first inequality. For code R, $P^R_T(\cdot \mid \operatorname{dom} D)$ is itself a semimeasure on answer lists, computable uniformly in $(T, \operatorname{dom} D)$ (Proposition A.8(a)), and the same argument gives the second. □

So the right line of slope 1 around a sufficient law (Corollary 5.5(a) of the companion paper on steeper priors) holds in both classes; in code Q without decidability it carries the flag cost n, which makes it useless, and for that reason the analyses of Section 1 use laws whose provability on the data is decidable.

A.3 Deterministic theories and precision

Proposition A.10 (Compatibility is monotone in logical strength)**.**

(a) Let $B \cup T$ be consistent and $\mathrm{Cn}(T) \supseteq \mathrm{Cn}(T')$. On every data set with which T' is compatible ($B \cup T'$ consistent and proving every datum), T is compatible. So among the deterministic hypotheses of W_A and M_A the posterior odds of T against T' equal their prior odds on every data set with which T' is compatible, and T' is incompatible whenever T is: data never move the odds towards the weaker theory.

(b) $T_0 \vdash T^\varepsilon$ for every $\varepsilon \geq 0$, so every data set compatible with the honest law T^ε is compatible with the exact law T_0, at every sample size and every precision. With prior weights $2^{-\|T\|}$ in a code in which the tolerance clause costs at least one bit, the posterior odds of T_0 against T^ε are at least $2^{\|T^\varepsilon\| - \|T_0\|} \geq 2$ on every such data set.

(c) If $\sup_t |e(t)| > 0$, then T_0 is false and, on a question at precision $\delta' < \sup_t |e(t)|$, proves a false answer at some times; T^ε with $\varepsilon \geq \sup_t |e(t)|$ is true and is silent wherever its band straddles a cell boundary.

Proof. A consistent theory that proves everything T' proves proves every datum that T' proves; this is (a), and (b) is (a) with $T = T_0$, $T' = T^\varepsilon$. (c) is immediate from the definitions. □

Proposition A.11 (Precision monotonicity)**.** *If $B \cup T$ proves "$q(t) \in [v - \delta', v + \delta']$", then it proves "$q(t) \in [v - \delta, v + \delta]$" for every $\delta \geq \delta'$. So compatibility with data at a fixed set of centres is monotone in the precision.*

Proof. The first interval is contained in the second. □

Proposition A.12 (Guaranteed tolerances)**.** *In regime G:*

(a) T_0 proves datum j iff $|e(t_j) - \zeta_j| \leq \delta$, and refutes it otherwise.

(b) If $\delta \geq \varepsilon^* + \delta_0$, then T_0 is compatible, at every n, with every such data set whose errors lie in the essential range (so almost surely).

(c) Suppose the pairs $(e(t_j), \zeta_j)$ are i.i.d., $e(t_j)$ is independent of ζ_j, both $\pm\delta_0$ lie in the support of the law of ζ_j, and $\delta < \varepsilon^* + \delta_0$. Then the per-datum refutation probability $r(\delta) := \mathbb{P}[|e - \zeta| > \delta]$ is positive, T_0 survives n data with probability $(1 - r(\delta))^n$, and it is almost surely refuted eventually.

So the switch point $\delta = \varepsilon^* + \delta_0$ is sharp.

Proof. (a) T_0 proves $q(t_j) = f(t_j) = v_j + e(t_j) - \zeta_j$. This point lies in $[v_j - \delta, v_j + \delta]$ iff $|e(t_j) - \zeta_j| \leq \delta$, and if it does not, T_0 proves the negation of the datum. (b) $|e - \zeta| \leq |e| + |\zeta| \leq \varepsilon^* + \delta_0$ almost surely. (c) Pick $\varepsilon' < \varepsilon^*$ and $s > 0$ with $\varepsilon' + \delta_0 - s > \delta$. Since ε^* is the essential supremum of $|e|$, $\mathbb{P}[e > \varepsilon'] > 0$ or $\mathbb{P}[e < -\varepsilon'] > 0$. In the first case $\mathbb{P}[\zeta < -\delta_0 + s] > 0$ (support), and independence gives $\mathbb{P}[e - \zeta > \delta] > 0$; the second case is symmetric, with $\zeta > \delta_0 - s$. Refutations of distinct data are independent, so survival has probability $(1 - r(\delta))^n \to 0$, and a refuted theory stays refuted. □

Proposition A.13 (Dithered rounding)**.** *In regime R, given $e(t_j)$, T_0 proves datum j with probability $(1-|e(t_j)|/\Delta)^+$ and refutes it otherwise. So T_0 survives n data with probability $(1-p_\Delta)^n$, its first refutation has a geometric index with mean $1/p_\Delta$, and it is almost surely refuted eventually iff $\mathbb{P}[e \neq 0] > 0$, for every Δ, however coarse. For $\Delta \geq \varepsilon^*$, $p_\Delta = \mathbb{E}|e|/\Delta$: the exact law survives about $\Delta/\mathbb{E}|e|$ data.*

Proof. T_0 proves "$q(t_j)$ lies in the cell of $f(t_j)$" and so refutes every other cell. Two points at distance $|e| < \Delta$ fall into different cells of a uniformly shifted grid of width Δ with probability $|e|/\Delta$, and at distance at least Δ they always do; this gives the conditional proof probability $(1 - |e(t_j)|/\Delta)^+$. The refutation indicators of distinct data are independent (they depend only on each datum's own error and dither), which gives the product $(1-p_\Delta)^n$ and the geometric law of the first refutation; and $p_\Delta > 0$ iff $\mathbb{P}[e \neq 0] > 0$. For $\Delta \geq \varepsilon^*$, $\min(|e|/\Delta, 1) = |e|/\Delta$ almost surely. □

This corrects the natural expectation that coarse data protect a false exact law: precision belongs to the data set, not to the datum. An exact law with a sinusoidal error of amplitude $1'$ is refuted by positions rounded to a dithered grid of 1° after about 94 data (the computations below), and Proposition A.14 compares several exact laws. We have not found these two regimes stated elsewhere; precision-dependent selection as such is familiar from accounts based on AIC, MDL and MML (cf. Forster and Sober, 1994; Rissanen, 1978; Wallace, 2005).

Proposition A.14 (Which exact theory a proof-requiring inducer settles on)**.** *Let $T_1, \ldots, T_k$ be exact pointwise laws for the same quantity, with lengths $\|T_1\| < \cdots < \|T_k\|$ and errors $e_1, \ldots, e_k$, and let the inducer choose the shortest compatible theory in this pool (the MAP of any prior that decreases with length, or K_A restricted to the pool). Let the data be i.i.d.*

(G) Under the hypotheses of Proposition A.12(c) at precision δ, almost surely the choice is eventually $T_{i^*(\delta)}$, $i^*(\delta) := \min\{i : \varepsilon_i^* + \delta_0 \leq \delta\}$; if no i qualifies, every theory is eventually refuted. As δ decreases past the thresholds $\varepsilon_i^* + \delta_0$, the eventual choice moves to longer and more exact theories.

(R) In regime R, for every $\Delta > 0$, almost surely the choice is eventually the shortest T_i with $\mathbb{P}[e_i \neq 0] = 0$, whatever Δ is; if there is none, every theory is eventually refuted. At sample size n, T_i survives with probability $(1 - p_{\Delta,i})^n$, and for $\Delta \geq \varepsilon_i^*$ it is abandoned after about $\Delta/\mathbb{E}|e_i|$ data.

Proof. Apply Proposition A.12 and Proposition A.13 to each T_i. The choice at n is the first survivor in order of length, and a finite pool has a last refutation time almost surely. □

Computations. In regime G, with $\varepsilon^* = 1'$, $\delta_0 = 0.5'$ and ζ uniform on $[-\delta_0, \delta_0]$, the refutation probability is exactly $r(\delta) = 0.218$ at $\delta = 1'$, 6.0×10^{-4} at $\delta = 1.49'$ and 0 for $\delta \geq 1.5'$ for a sinusoidal error, and 0.025, 5.0×10^{-4} and 0 for an error equal to $\pm\varepsilon^*$ on 5% of the times and 0 otherwise: positive below $\varepsilon^* + \delta_0$ and zero above it for both. In regime R, simulated survival after 1, 10 and 100 data matches $(1 - p_\Delta)^n$ at every Δ from $0.5'$ to $600'$; for the sinusoid with $\varepsilon^* = 1'$ the mean first-refutation index is 94.1 against the exact $1/p_\Delta = \pi\Delta/2 = 94.25$ at $\Delta = 60'$, and 942.7 against 942.48 at $\Delta = 600'$. For the pool T_1 (length 100, $\varepsilon^* = 10'$), T_2 (length 150, $\varepsilon^* = 1'$), T_3 (length 400, exact), with sinusoidal errors and 2000 runs, every run chooses the predicted theory at $n = 10^4$ in regime G with $\delta_0 = 0.5'$ at each of $\delta = 20'$, $10.5'$, $5'$, $1.5'$, $1.2'$ (predictions T_1, T_1, T_2, T_2, T_3); in regime R every run ends at T_3, T_1 being abandoned after about 94 data at $\Delta = 10^\circ$ and 9.4 at $\Delta = 1^\circ$, and T_2 after about 942 and 94.

A.4 Honest laws, laws with exceptions, laws with error models

Forms and pricings. Work in regime R. The forms are: the empty law $\emptyset$; the *patched tolerance law* $T^{\varepsilon,-Z}$, the tolerance law T^{ε} withdrawn at the set Z of times of the data it refutes (fewer patches are inconsistent with the data, more only add cost); for $\varepsilon = 0$ the exact law, written T_0 while it has no patch and T_0^{-Z} after; for $\varepsilon \geq \varepsilon^*$ the honest law T^{ε}, which is true and never patched; and a true exact theory T_{tr}. Their lengths, relative to a common part, are $\|\emptyset\| = 0$, $\|T_0\| = \ell_0$, $\|T^{\varepsilon}\| = \ell_0 + h$ (h the length of the tolerance clause and its numeral) and $\|T_{\mathrm{tr}}\| = \ell_0 + h_{\mathrm{tr}}$. A pricing fixes the price γ_{si} of a datum on which a form is silent and the price γ_{ex} of an exception:

- *Q/det*: code Q, each patch naming its time inside the law at c_{time} bits; $\gamma_{\mathrm{si}} = \gamma$ and $\gamma_{\mathrm{ex}} = \lambda c_{\mathrm{time}} + \gamma$. This is a finite-horizon cost model (Proposition A.17).
- *Q/rand*: code Q with randomised exceptions: a theory machine withdraws the law at each time independently with probability ρ, and the fit is charged $-\log(1-\rho)$ per kept datum and $-\log\rho + \gamma$ per exception. The share ρ can be stated in $O(\log n)$ bits or integrated out: drawing ρ uniformly from the tape and then withdrawing independently is a single theory machine, whose cost for k exceptions among n data is $\log\left((n+1)\binom{n}{k}\right) + k\gamma$. With $\rho = p_\Delta$, or integrated out, the rate is $H_2(p_\Delta) + p_\Delta\gamma$, where H_2 is the binary entropy, so $\gamma_{\mathrm{ex}} = \gamma + H_2(p_\Delta)/p_\Delta$. These are exact theory-machine costs for every n.
- *R/det* and *R/rand*: as above, with a silent datum priced by random completion inside the band (code R); an exception of a patched law is priced γ in code R too (Proposition A.4(c)).

SAI's pricing is R/rand: the honest law with random completion and the exact law with random exceptions are theory machines, and R/rand compares their weights in M_A's mixture.

Lemma A.15 (What a tolerance law does with one datum)**.** *In regime R, fix a datum with error e and let $\varepsilon \geq 0$. Over the dither, T^{ε} refutes the datum with probability* $\min\left((|e|-\varepsilon)^+/\Delta, 1\right)$*, proves it with probability* $\left(1-(\varepsilon+\max(|e|,\varepsilon))/\Delta\right)^+$*, and is silent otherwise. In the* coarse regime $\Delta \geq \varepsilon + \max(\varepsilon, \varepsilon^*)$ *it is silent with probability exactly $2\varepsilon/\Delta$, whatever e is, and refutes with probability $(|e|-\varepsilon)^+/\Delta$. For $\varepsilon = 0$ it refutes with probability* $\min(|e|/\Delta, 1)$ *and is never silent.*

Proof. Place q at 0. The cell containing it is $[-u\Delta, (1-u)\Delta)$ with u uniform on $[0,1)$, and the band is $[e-\varepsilon, e+\varepsilon]$, since $f = q + e$; by symmetry let $e \geq 0$. *Refutation* means that the band misses the cell. If $e \leq \varepsilon$, the band contains 0, so this is impossible. If $e > \varepsilon$, the band lies in $(0,\infty)$, and it misses the cell iff $(1-u)\Delta \leq e - \varepsilon$, of probability $\min((e-\varepsilon)/\Delta, 1)$. *Proof of the datum* means that the band lies in the cell: $-u\Delta \leq e - \varepsilon$ and $e + \varepsilon < (1-u)\Delta$. If $e \leq \varepsilon$, this is $u \in [(\varepsilon - e)/\Delta, (\Delta - \varepsilon - e)/\Delta)$, of probability $(1 - 2\varepsilon/\Delta)^+$; if $e > \varepsilon$, the first condition holds automatically and the second is $u < (\Delta - e - \varepsilon)/\Delta$. Both cases equal $(1 - (\varepsilon + \max(e,\varepsilon))/\Delta)^+$. When $\Delta \geq \varepsilon + \max(\varepsilon, \varepsilon^*)$ the positive parts are inactive for every e in the essential range, and silence has probability $1 - (1 - (\varepsilon + \max(e,\varepsilon))/\Delta) - (\max(e,\varepsilon) - \varepsilon)/\Delta = 2\varepsilon/\Delta$. □

The coarse-regime condition cannot be weakened to $\Delta \geq \varepsilon + \varepsilon^*$ when $\varepsilon > \varepsilon^*$: with $|e| = \varepsilon^* = 1$, $\varepsilon = 2$ and $\Delta = 3$ the formula $2\varepsilon/\Delta$ would give a silence probability of $4/3$.

Proposition A.16 (Rates)**.** *In regime R, write $z_j := (e_j, u_j)$. Each form X has $J_\lambda(X; D_n) = a_X + \sum_{j \leq n} c_X(z_j)$ with a constant a_X and i.i.d. per-datum costs $c_X(z_j)$ (under Q/rand and R/rand with ρ stated; integrating ρ out adds $O(\log n)$). In the coarse regime $\Delta \geq 2\varepsilon^*$ the rates $m_X := \mathbb{E}c_X$ are: $m_\emptyset = \gamma$; $m_{T_{\mathrm{tr}}} = 0$; $m_{T_0} = \infty$ if $\mathbb{P}[e \neq 0] > 0$ (an unpatched exact law is eventually refuted); $m_{T_0^{-z}} = p_\Delta\gamma_{\mathrm{ex}}$, that is $p_\Delta(\lambda c_{\mathrm{time}} + \gamma)$ under the det pricings and $H_2(p_\Delta) + p_\Delta\gamma$ under the rand pricings, with $p_\Delta = \mathbb{E}|e|/\Delta$; and $m_{T^{\varepsilon^*}} = s_\Delta\gamma_{\mathrm{si}}$ with $s_\Delta = 2\varepsilon^*/\Delta$, that is $s_\Delta\gamma$ in code Q and $s_\Delta\beta_e$ in code R, β_e as in Lemma A.20.*

Proof. For the patched exact law at its cheapest patch set, a datum costs γ_{ex} if T_0 refutes it and 0 otherwise: under det the patch names the time ($\lambda c_{\mathrm{time}}$) and the datum is then literal (γ); under rand it is the flag code. Lemma A.15 with $\varepsilon = 0$ gives the probability p_Δ. The honest law costs γ_{si} on a datum iff it is silent, which in the coarse regime has probability $2\varepsilon^*/\Delta$ independently of e (Lemma A.15); in code R the expected cost given silence, averaged over the errors, is β_e (Lemma A.20(i)). □

Proposition A.17 (The price of naming exceptions deterministically)**.** *In regime R, let $Z_n \subseteq \{1, \dots, n\}$ be the indices of the data that the exact law refutes, so that the events $j \in Z_n$ are independent with probability $p_\Delta \in (0, 1)$.*

(a) For every prefix-free code of the finite subsets of $\{1, \dots, n\}$, even one that depends on n, the expected code length of Z_n is at least $nH_2(p_\Delta)$; so the expected cost per exception is at least $H_2(p_\Delta)/p_\Delta$ bits, the cost of randomised exceptions.

(b) If the patched law names each patched time t_j by a numeral in a prefix-free code ℓ_{num} of numerals, then $\mathbb{E}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j) \geq p_\Delta n \log n$, so the cost per exception is at least $\log n$ on average. If $\ell_{\mathrm{num}}(t_j) \geq \log j - c$ for all j (any standard numeral code), the cost per datum $\frac{1}{n}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j)$ tends to ∞ almost surely.

(c) Hence Q/det and R/det with a constant c_{time} are cost models valid only while the times used cost about c_{time} bits to name, that is for $n \lesssim 2^{c_{\mathrm{time}}}$. Asymptotically, at fresh times, deterministic patching costs at least as much per exception as randomised exceptions, and literal patching costs unboundedly more; so the patched exact law under det is eventually beaten by the exact law with random exceptions.

Proof. (a) Let P_Z be the law of Z_n and ℓ' a prefix-free code. By Kraft's inequality $\sum_Z 2^{-\ell'(Z)} \leq 1$, and by Jensen's inequality, with H the entropy,

$$\mathbb{E}\ell'(Z_n) - H(Z_n) = -\sum_Z P_Z(Z) \log \frac{2^{-\ell'(Z)}}{P_Z(Z)} \geq -\log \sum_Z 2^{-\ell'(Z)} \geq 0,$$

where $H(Z_n) = nH_2(p_\Delta)$. The expected number of exceptions is $p_\Delta n$. (b) The numerals $t_1, \dots, t_n$ are distinct, so Kraft gives $\sum_{j\leq n} 2^{-\ell_{\mathrm{num}}(t_j)} \leq 1$, and Jensen gives $2^{-\frac{1}{n}\sum_j \ell_{\mathrm{num}}(t_j)} \leq \frac{1}{n}\sum_j 2^{-\ell_{\mathrm{num}}(t_j)} \leq \frac{1}{n}$, that is $\sum_{j\leq n} \ell_{\mathrm{num}}(t_j) \geq n \log n$. Since $\mathbb{P}[j \in Z_n] = p_\Delta$ for every j, $\mathbb{E}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j) \geq p_\Delta n \log n$. If $\ell_{\mathrm{num}}(t_j) \geq \log j - c$, then $\frac{1}{n}\sum_{j\in Z_n} \ell_{\mathrm{num}}(t_j) \geq \frac{1}{n}\#\{j \in Z_n : j > n/2\}(\log n - 1 - c)$, and the count over n tends to $p_\Delta/2$ almost surely by the strong law of large numbers. (c) follows from (a), (b) and Proposition A.16. □

A computation (code in the accompanying repository) gives, at $p_\Delta = 0.01$, 15.0, 23.0 and 30.3 bits per exception for literal listing by Elias delta numerals at $n = 10^3$, 10^5 and 10^7, and 9.02, 8.13 and 8.08 bits for an enumerative code of the set, against $H_2(p_\Delta)/p_\Delta = 8.08$.

Proposition A.18 (The peakedness criterion)**.** *Let $\Delta \geq 2\varepsilon^*$, under any of the four pricings (the deterministic ones as a cost model).*

(a) The honest law T^{ε^*} has a lower rate than the exact law with exceptions T_0^{-Z} iff $\mathrm{pk}(e) < \gamma_{\mathrm{ex}}/(2\gamma_{\mathrm{si}})$.

(b) If the exact law with exceptions comes from a law f_1 with error e_1, and the honest law from a law f_2 of the same length up to $O(1)$, with tolerance $\varepsilon_2 \geq \operatorname{ess\,sup}|e_2|$ and $\Delta \geq 2\varepsilon_2$, then the honest law has the lower rate iff $\varepsilon_2/\mathbb{E}|e_1| < \gamma_{\mathrm{ex}}/(2\gamma_{\mathrm{si}})$.

Proof. By Proposition A.16, $s_\Delta \gamma_{\mathrm{si}} < p_\Delta \gamma_{\mathrm{ex}}$ iff $2\varepsilon^* \gamma_{\mathrm{si}} < \mathbb{E}|e|\,\gamma_{\mathrm{ex}}$. In (b), $s_\Delta = 2\varepsilon_2/\Delta$ (Lemma A.15) and $p_\Delta = \mathbb{E}|e_1|/\Delta$. □

Under Q/det the criterion reads $\lambda c_{\mathrm{time}} > \gamma(2\mathrm{pk}(e) - 1)$. Simulated rates for a sinusoid ($\mathrm{pk} = \pi/2$) with $c_{\mathrm{time}} = 22$ pick the exact law with exceptions at $\lambda = 1$ and the honest law at $\lambda = 2$ for $\Delta = 2'$, $3'$ and $10'$, and the honest law at both for $\Delta = 60'$, as the threshold $(\lambda c_{\mathrm{time}} + \gamma)/(2\gamma)$ predicts; for an error with $\mathrm{pk} = 20$ they pick the exact law with exceptions everywhere. Under Q/rand it reads $\mathrm{pk}(e) < \frac{1}{2} + H_2(p_\Delta)/(2p_\Delta\gamma)$; since $H_2(p)/p \approx \log(\mathrm{e}/p)$ for small p, the threshold grows only logarithmically in $1/p_\Delta$: at $\gamma = \log 360$ it is 1.17 at $p_\Delta = 10^{-3}$, 1.76 at 10^{-6} and 2.93 at 10^{-12}. For the sinusoid it is 0.637, 0.742 and 0.971 at $\Delta = 3'$, $10'$ and $60'$, so the exact law with random exceptions wins although the error is spread. Within the pool $\{\emptyset, T_0^{-Z}, T^{\varepsilon^*}\}$ under R/rand the threshold is $(\gamma + H_2(p_\Delta)/p_\Delta)/(2\beta_e)$, roughly 5 to 30 for typical cell widths (9.2–9.3 for the sinusoid, 20.4–20.5 for the error with $\mathrm{pk} = 20$). This comparison is relative to its pool: the exact law with random exceptions completes its exceptions over the whole circle, while the equally short two-point completion $\mathrm{HY}(\rho)$ completes them inside the band, and against it the error with $\mathrm{pk} = 20$ is held as "exact except in about 10% of cases" at a third of the uniform band's rate (Proposition 1.2(d) and the computation after its proof).

Proposition A.19 (Optimal tolerance)**.** *Under Q/det in the regime $\Delta \geq 2\varepsilon^*$, the patched tolerance law $T^{\varepsilon,-Z}$ with $0 \leq \varepsilon \leq \varepsilon^*$ has rate*

$$\bar{m}(\varepsilon) = \frac{2\varepsilon\gamma + (\lambda c_{\mathrm{time}} + \gamma)\,\mathbb{E}(|e| - \varepsilon)^+}{\Delta}.$$

$\bar{m}$ is convex, with right derivative $\big(2\gamma - (\lambda c_{\mathrm{time}} + \gamma)\mathbb{P}[|e| > \varepsilon]\big)/\Delta$, and its minimisers on $[0, \varepsilon^*]$ are the ε with $\mathbb{P}[|e| > \varepsilon] \leq \theta_{\mathrm{ex}}$ and, if $\varepsilon > 0$, $\theta_{\mathrm{ex}} \leq \mathbb{P}[|e| \geq \varepsilon]$, where $\theta_{\mathrm{ex}} := 2\gamma/(\lambda c_{\mathrm{time}} + \gamma)$. So the optimal law states the $(1 - \theta_{\mathrm{ex}})$-quantile of $|e|$ as its tolerance and lists the remaining share θ_{ex} of cases as exceptions. If $\theta_{\mathrm{ex}} \geq 1$ ($\lambda c_{\mathrm{time}} \leq \gamma$), the exact law with patches is optimal. The pure honest law ($\varepsilon = \varepsilon^*$, no exceptions) is optimal iff $\mathbb{P}[|e| \geq \varepsilon^*] \geq \theta_{\mathrm{ex}}$, which never holds for an error law without an atom at its maximum.

Proof. A datum costs γ if it is not proved, and in addition $\lambda c_{\mathrm{time}}$ if it is refuted (the patch). By Lemma A.15, with $\Delta \geq 2\varepsilon^* \geq \varepsilon + \varepsilon^*$, $\mathbb{P}[\text{not proved}] = (2\varepsilon + \mathbb{E}(|e| - \varepsilon)^+)/\Delta$ and $\mathbb{P}[\text{refuted}] = \mathbb{E}(|e| - \varepsilon)^+/\Delta$, which gives $\bar{m}$. The function $\varepsilon \mapsto \mathbb{E}(|e| - \varepsilon)^+$ is convex, with right derivative $-\mathbb{P}[|e| > \varepsilon]$ and left derivative $-\mathbb{P}[|e| \geq \varepsilon]$. This gives convexity and the derivatives of $\bar{m}$, and a convex function is minimised exactly where its right derivative is non-negative and its left derivative non-positive (at $\varepsilon = 0$ only the right derivative matters), which is the stated condition. The last two claims are the cases $\varepsilon = 0$ and $\varepsilon = \varepsilon^*$ of it. □

Minimising $\theta\varepsilon + \mathbb{E}(|e| - \varepsilon)^+$ over ε is the quantile-loss fact, and in ν-support-vector regression the parameter ν likewise bounds the share of points outside an ε-tube whose width is a quantile of the residuals (cf. Schölkopf et al., 2000). What is specific here is the reading for axiom inducers: the tube is a tolerance clause, the points outside it are patched exceptions, and θ_{ex} is a ratio of literal costs. The steeper the prior, or the dearer the naming of an exception, the smaller θ_{ex}: the law states a wider tolerance and lists fewer exceptions. A computation at $\Delta = 10'$, $c_{\mathrm{time}} = 22$ finds the grid minimiser at 0.52, 0.80 and 0.94 for a sinusoid of amplitude 1 at $\lambda = 1, 2, 4$, against the predicted quantiles 0.496, 0.807 and 0.939.

Lemma A.20 (The code-R cost of the honest law, exactly)**.** *Let $\varepsilon \geq \varepsilon^*$ and let $w := (\varepsilon - e)/(2\varepsilon) \in [0,1]$ be the position of the true value in the band. Put $\vartheta(z) := 1 - z + z\ln z$ for $0 \leq z < 1$ ($\vartheta(0) = 1$) and $\vartheta(z) := 0$ for $z \geq 1$. The expected code-R cost of a datum for T^ε, over the dither, is*

$$\bar{c}_\Delta(w) = \begin{cases} \dfrac{2\varepsilon}{\Delta}\big(\log \mathrm{e} - H_2(w)\big) & \text{if } \Delta \geq 2\varepsilon, \\ \log\dfrac{2\varepsilon}{\Delta} + \log \mathrm{e}\,\Big[\vartheta\Big(\dfrac{2\varepsilon w}{\Delta}\Big) + \vartheta\Big(\dfrac{2\varepsilon(1-w)}{\Delta}\Big)\Big] & \text{if } \Delta \leq 2\varepsilon. \end{cases}$$

(i) In the coarse regime the cost given silence is $\log \mathrm{e} - H_2(w)$. Averaged over the errors, $\gamma_{\mathrm{si}} = \beta_e := \log \mathrm{e} - \mathbb{E}H_2(w) \in [\log \mathrm{e} - 1, \log \mathrm{e}] = [0.443, 1.443]$ bits.

(ii) In the fine regime T^ε is silent on every datum, and $\log(2\varepsilon/\Delta) \leq \bar{c}_\Delta(w) \leq \log(2\varepsilon/\Delta) + \log \mathrm{e}$; the upper bound is approached as $w \to 0$ at $\Delta = 2\varepsilon$.

(iii) For every w, $\bar{c}_\Delta(w)$ is continuous and strictly decreasing in $\Delta \in (0, \infty)$.

Proof. Coarse regime. The band straddles a cell boundary with probability $2\varepsilon/\Delta$ (Lemma A.15, since $\varepsilon \geq \varepsilon^*$); otherwise it lies in a cell and the datum costs 0. Given a straddle, exactly one boundary lies in the band, at a relative position v uniform on $[0,1]$ (a uniformly shifted lattice of spacing at least the band's length). The true cell contains the part of the band on the true value's side, of relative length v if $w < v$ and $1 - v$ otherwise, and random completion inside the band puts that much probability on it. So the expected cost given a straddle is, in nats,

$$\int_w^1 -\ln v \, dv + \int_0^w -\ln(1-v)\, dv = 1 + w\ln w + (1-w)\ln(1-w),$$

which is $\log \mathrm{e} - H_2(w)$ in bits.

Fine regime. Let y be the length of the part of the true cell inside the band; the cost is $\log(2\varepsilon/y) = \log(2\varepsilon/\Delta) - \log(y/\Delta)$. With $a := 2\varepsilon w/\Delta$ and $b := 2\varepsilon(1-w)/\Delta$, so that $a + b = 2\varepsilon/\Delta \geq 1$, the band is $[-a\Delta, b\Delta]$ around the true value, and with dither u the true cell is $[-u\Delta, (1-u)\Delta)$, so $y/\Delta = \min(u, a) + \min(1-u, b)$. If $a, b \geq 1$, then $y = \Delta$ and the excess $-\ln(y/\Delta)$ is 0. If $a < 1 \leq b$, then $y/\Delta = 1$ for $u < a$ and $1 + a - u$ for $u \geq a$, so the expected excess is $\int_a^1 -\ln(1 + a - u)\, du = \int_a^1 -\ln s \, ds = \vartheta(a)$; the case $b < 1 \leq a$ is symmetric. If $a, b < 1$, then $1 - b \leq a$, and $y/\Delta = u + b$ for $u < 1 - b$, 1 on $[1-b, a]$, and $a + 1 - u$ for $u > a$; the expected excess is $\int_b^1 -\ln s \, ds + \int_a^1 -\ln s \, ds = \vartheta(a) + \vartheta(b)$.

(i) is the coarse case, and $\mathbb{E}H_2(w) \in [0,1]$. (ii) In the fine regime $2\varepsilon \geq \Delta$, so T^ε is silent on every datum by Lemma A.15. $\vartheta \geq 0$ and ϑ is decreasing on $[0,1]$ ($\vartheta'(z) = \ln z$). In the last case $\vartheta(a) + \vartheta(b)$ is largest on $a + b = 1$, where it equals $1 - H_2(a)\ln 2 \leq 1$; in the middle case $\vartheta(a) \leq \vartheta(0) = 1$. The value 1 is approached as $a \to 0$, $b \to 1$. (iii) Write $r := 2\varepsilon/\Delta$ and the cost in nats. In the coarse regime ($r \leq 1$) it is $r(1 - H_2(w)\ln 2)$, increasing in r because $1 - H_2(w)\ln 2 \geq 1 - \ln 2 > 0$. In the fine regime ($r \geq 1$) it is $\ln r + \vartheta(wr) + \vartheta((1-w)r)$, and on each piece its derivative in r is $\frac{1}{r}(1 + \sum z\ln z)$, the sum over $z \in \{wr, (1-w)r\}$ with $z < 1$; since $z\ln z \geq -1/\mathrm{e}$, this is at least $\frac{1}{r}(1 - 2/\mathrm{e}) > 0$. The cost is continuous in r (ϑ is continuous at 1), and at $r = 1$ the two forms agree, since $\vartheta(w) + \vartheta(1-w) = 1 - H_2(w)\ln 2$. So it is strictly increasing in r, that is strictly decreasing in Δ. □

A computation (code in the accompanying repository) confirms the closed form against a 5000-point quadrature over the dither to 10^{-4} bits on 36090 cells, with largest fine-regime excess $1.442695 = \log \mathrm{e}$; the coarse averages are $\beta_e = 0.885$ for a sinusoid and 0.493 for an error equal to $\pm\varepsilon^*$ on 5% of the times.

Band-confined completions. The machine $\mathrm{BC}_{\mathcal{G}}$ of Section 1.3 draws $\hat{w}_i \sim \mathcal{G}$ by the inverse distribution function from the uniform variables U_i of the proof of Proposition A.4, and emits the value $f(t_i) - \varepsilon + 2\varepsilon\hat{w}_i$ by narrowing intervals for the continuous part of $\mathcal{G}$ and exactly at an atom. A computable $\mathcal{G}$ is, for instance, piecewise linear with rational knots plus finitely many atoms at rational points. The argument of Proposition A.4(a) gives $K(\mathrm{BC}_{\mathcal{G}}) \le \|T^{\varepsilon}\| + K(\mathcal{G}) + c$, the random theory is always consistent, and on data at fresh times $\nu_{\mathrm{BC}_{\mathcal{G}}}(D) = \prod_j \mathcal{G}(C'_j)$ (cell endpoints are null events for continuous dithers).

Proof of Proposition 1.2. (a) As in Lemma A.20: a cell boundary falls into the band with probability $2\varepsilon/\Delta$, at a relative position v uniform on $[0,1]$ and independent of w. With no boundary in the band the cell contains the band and the datum costs 0. With a boundary at v, the true cell contains $[0,v)$ if $w < v$, of $\mathcal{G}$-mass $\mathcal{G}(v)$, and $[v,1]$ if $w > v$, of mass $1 - \mathcal{G}(v)$; this gives $c_{\mathcal{G}}$, which does not depend on Δ.

(b) Given v, the event $w < v$ has probability $\mathcal{F}(v)$, so the conditional expected cost is $\mathcal{F}(v)(-\log \mathcal{G}(v)) + (1-\mathcal{F}(v))(-\log(1-\mathcal{G}(v))) \ge H_2(\mathcal{F}(v))$, with equality iff $\mathcal{G}(v) = \mathcal{F}(v)$ (Gibbs' inequality). Integrate over v. The uniform band has $\mathcal{G}(v) = v$, which equals $\mathcal{F}$ on $[0,1]$ iff w is uniform, that is iff e is uniform on $[-\varepsilon, \varepsilon]$.

(c) Let $P^{\star}$ be the true conditional law of the answer list given the question list. The errors are independent of the questions, and $q = f - e$, so the conditional probability of the observed cell of $q(t_j)$ is $\mathcal{F}(C'_j)$; hence $P^{\star}(D_n) = \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n) = \prod_{j \le n} \mathcal{F}(C'_j)$, and $-\log \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n)$ is a sum of i.i.d. terms with finite mean $r_{\mathcal{F}}(\Delta)$; the strong law of large numbers gives the limit. Let G' be any theory machine. Answers to one question are pairwise B-incompatible, so for a fixed question list the events "the random theory is consistent and proves the answer list y" are disjoint across y; hence $\sum_y \nu_{G'}(y) \le 1$ and $\mathbb{E}_{P^{\star}}[\nu_{G'}(D_n)/P^{\star}(D_n) \mid \text{questions}] \le 1$. By Markov's inequality $\mathbb{P}[-\log \nu_{G'}(D_n) < -\log P^{\star}(D_n) - 2\log n] \le n^{-2}$, and by the Borel–Cantelli lemma, almost surely $-\log \nu_{G'}(D_n) \ge -\log P^{\star}(D_n) - 2\log n$ for all large n. Dividing by n gives the lim inf bound, and $G' = G_U$ gives it for L_A. If $\mathcal{F}$ is computable, dominance gives $L_A(D_n) \le K(\mathrm{BC}_{\mathcal{F}}) + c - \log \nu_{\mathrm{BC}_{\mathcal{F}}}(D_n)$, hence the limit. In the coarse regime $r_{\mathcal{F}}(\Delta) = (2\varepsilon/\Delta)c_{\mathcal{F}}$ by (a) and the equality case of (b).

(d) Write $\sigma := 1 - \rho$ for the weight of the atom at $\frac{1}{2}$. The family is symmetric, $\mathcal{G}(1-v) = 1 - \mathcal{G}(v)$, so the cost given w equals the cost given $1-w$, and we may take $w \le \frac{1}{2}$. *Convexity.* For fixed (w,v) the probability assigned to the true side is affine in σ and positive for $\sigma \in [0,1)$, and $-\ln$ of a positive affine function is convex; so $c_{\mathrm{HY}(\rho)}$, an expectation of convex functions, is convex in σ, hence in $\rho \in (0,1]$. *Right derivative at $\sigma = 0$, in nats.* For $v < w$ the true side has probability $1 - (1-\sigma)v$, and the derivative of $-\ln$ of it at $\sigma = 0$ is $-v/(1-v) \in [-1,0]$; for $w < v < \frac{1}{2}$ the probability is $(1-\sigma)v$, with derivative $+1$; for $v > \frac{1}{2}$ it is $\sigma + (1-\sigma)v$, with derivative $-(1-v)/v \in [-1,0]$. Difference quotients of convex functions decrease monotonically as $\sigma \downarrow 0$, so by monotone convergence the derivative of the expectation is the expectation of the derivatives:

$$c'(0^+) = \mathbb{E}\Big[-\int_0^w \tfrac{v}{1-v}\,dv + (\tfrac{1}{2} - w) - \int_{1/2}^1 \tfrac{1-v}{v}\,dv \Big] = \mathbb{E}\big[\ln(1-w) + 1 - \ln 2\big],$$

using $\int_0^w \frac{v}{1-v}dv = -\ln(1-w) - w$ and $\int_{1/2}^1 \frac{1-v}{v}dv = \ln 2 - \frac{1}{2}$. For $w \le \frac{1}{2}$, $1 - w = (1 + |e|/\varepsilon)/2$, so $c'(0^+) = \mathbb{E}\ln(1 + |e|/\varepsilon) + 1 - 2\ln 2$. *The criterion.* If $c'(0^+) < 0$, some $\sigma > 0$ beats $\sigma = 0$, which is the uniform band; if $c'(0^+) \ge 0$, convexity gives $c(\sigma) \ge c(0)$ for every σ. *Necessity.* $\ln(1+z) \ge z\ln 2$ on $[0,1]$ (a concave function lies above its chord), so $\mathbb{E}\ln(1+|e|/\varepsilon) \ge \ln 2/\mathrm{pk}_{\varepsilon}(e)$, and $c'(0^+) < 0$ forces $\mathrm{pk}_{\varepsilon}(e) > \ln 2/(2\ln 2 - 1)$. *Sufficiency.* Jensen's inequality gives

$\mathbb{E}\ln(1+|e|/\varepsilon) \le \ln(1+1/\mathrm{pk}_\varepsilon(e))$, which is below $2\ln 2-1$ iff $\mathrm{pk}_\varepsilon(e) > \mathrm{e}/(4-\mathrm{e})$. *Boundary.* $\int_0^1 \ln(1+z)\,dz = 2\ln 2 - 1$. □

The following computation (synthetic errors, 2×10^5 samples, band $\varepsilon = \varepsilon^*$; code in the accompanying repository) gives per-straddle costs in bits in the coarse regime. SH is a 40-bin smoothed histogram fitted on a random half of the samples and evaluated on the other half, $c_{\mathcal{F}}$ is the bound of Proposition 1.2(b) on that half, and $c'(0^+)$ (in nats) is the derivative that decides whether some HY(ρ) with $\rho < 1$ beats the uniform band.

error	pk	uniform	best HY	ρ	SH	$c_{\mathcal{F}}$	$c'(0^+)$
sinusoid	1.57	0.8854	0.8854	1	0.8514	0.8510	$+0.087$
$\pm\varepsilon^*$ on 5%	20.1	0.4924	0.1790	0.099	0.2006	0.1704	-0.352
$\lvert e\rvert$ uniform	2.00	0.7213	0.7213	1	0.7227	0.7227	≈ 0
0.25 or 1 (90/10%)	3.08	0.5832	0.5543	0.68	0.4797	0.4640	-0.116

So the two-point form beats the uniform band exactly when $c'(0^+) < 0$, and the fitted histogram comes within about 0.02 bits of the Gibbs bound of the half it did not see.

Proposition A.21 (Fine data)**.** *Let $\Delta \le 2\varepsilon^*$. The honest law T^{ε^*} is silent on every datum.*

(a) In code Q its rate is $\gamma = m_\emptyset$, so it never beats the empty law, and T_0^{-Z} beats the empty law iff $p_\Delta \gamma_{\mathrm{ex}} < \gamma$.

(b) In code R its rate is at most $\log(2\varepsilon^*/\Delta) + \log \mathrm{e}$, which is below $m_\emptyset = \gamma = \log(\mathsf{R}/\Delta)$ by at least $\log(\mathsf{R}/2\varepsilon^*) - \log \mathrm{e}$ bits per datum; this is positive iff $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$, about 66.2° for a longitude.

(c) If moreover $\varepsilon^* < \mathsf{R}/(2\mathrm{e})$ and $\mathbb{P}[|e| < \Delta]\log(1/\Delta) \to 0$ as $\Delta \to 0$ (for instance, if $|e|$ has a bounded density near 0), then in code R the honest law has the least rate among $\emptyset$, T_0^{-Z} and T^{ε^*} for all small Δ.

Proof. Silence and (a) are Lemma A.15 and Proposition A.16's argument with silence probability 1. (b) is Lemma A.20(ii). (c) $m_{T_0^- z} \ge p_\Delta \gamma$, and $1 - p_\Delta = \mathbb{E}(1-|e|/\Delta)^+ \le \mathbb{P}[|e| < \Delta]$, so $m_{T_0^- z} - m_{T^{\varepsilon^*}} \ge \log(\mathsf{R}/2\varepsilon^*) - \log \mathrm{e} - \mathbb{P}[|e| < \Delta]\log(\mathsf{R}/\Delta)$, which is positive for small Δ; and (b) handles the empty law. □

So under SAI's pricing an approximate law keeps its value at precisions finer than its error: it narrows each answer to its band, which is worth $\log(\mathsf{R}/2\varepsilon^*)$ bits per datum (about 13 bits for a $1'$ error on a full circle of longitude), and by Proposition 1.2 a completion shaped like the error narrows it further.

Theorem A.22 (Almost-sure selection)**.** *Let $\mathcal{X}$ be a finite set of forms with $J_\lambda(X; D_n) = a_X + \sum_{j\le n} c_X(z_j) + r_X(n)$, where the z_j are i.i.d., $c_X(z) \in [0,\infty]$, and $r_X(n) = o(n)$ almost surely ($r_X = 0$ for deterministic forms and for stated ρ; $O(\log n)$ when ρ is integrated out). Let $m_X := \mathbb{E}c_X \in [0,\infty]$, $m_{\min} := \min_X m_X < \infty$ and $\mathcal{X}^* := \{X : m_X = m_{\min}\}$.*

(i) Almost surely, for all large n every λ-MAP within $\mathcal{X}$ lies in $\mathcal{X}^*$. If $\mathcal{X}^* = \{X^*\}$, the λ-MAP is eventually X^* alone.

(ii) Let $X, Y \in \mathcal{X}^*$ with $r_X = r_Y = 0$. If $c_X - c_Y$ is almost surely finite, with finite variance, and not almost surely constant, then almost surely $J_\lambda(X; D_n) < J_\lambda(Y; D_n)$ for infinitely many n and $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$ for infinitely many n: there is no eventual winner. If $c_X - c_Y = 0$ almost surely, a_X against a_Y decides at every n.

(iii) Let $r_X = r_Y = 0$, $m_Y < m_X < \infty$, $\mathbb{E}|c_X - c_Y| < \infty$ and $x := a_Y - a_X > 0$. Let τ_x be the first n with $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$, ς_x the last n with $J_\lambda(Y; D_n) \geq J_\lambda(X; D_n)$, and $\mu := m_X - m_Y$. Then $\mathbb{E}\tau_x > x/\mu$; if moreover $c_X - c_Y \leq b$ almost surely, then $\mathbb{E}\tau_x \leq (x+b)/\mu$; and on almost every sample path, $\tau_x/x \to 1/\mu$ and $\varsigma_x/x \to 1/\mu$ as $x \to \infty$.

Proof. The probabilistic facts used are standard (see Durrett, 2019). (i) Let $X^* \in \mathcal{X}^*$; then $\frac{1}{n} J_\lambda(X^*; D_n) \to m_{\min}$ almost surely. Let $X \notin \mathcal{X}^*$. If $\mathbb{P}[c_X = \infty] > 0$, almost surely some datum has infinite cost, after which $J_\lambda(X; D_n) = \infty$. If $c_X < \infty$ almost surely and $m_X = \infty$, then for each K, $\frac{1}{n}\sum_{j \leq n} \min(c_X(z_j), K) \to \mathbb{E}\min(c_X, K)$, and $\mathbb{E}\min(c_X, K) \to \infty$ as $K \to \infty$, so $\frac{1}{n} J_\lambda(X; D_n) \to \infty$. If $m_X < \infty$, then $\frac{1}{n}(J_\lambda(X; D_n) - J_\lambda(X^*; D_n)) \to m_X - m_{\min} > 0$. Since $\mathcal{X}$ is finite, the claim follows. (ii) $S_n := \sum_{j \leq n}(c_X - c_Y)(z_j)$ is a random walk with mean-zero, non-degenerate increments of finite variance. For each K, $\mathbb{P}[S_n > K] \to \frac{1}{2}$ by the central limit theorem, so $\mathbb{P}[S_n > K \text{ infinitely often}] \geq \limsup_n \mathbb{P}[S_n > K] = \frac{1}{2}$. This event is invariant under finite permutations of the increments, so by the Hewitt–Savage 0–1 law its probability is 1. Intersecting over $K \in \mathbb{N}$ gives $\limsup_n S_n = +\infty$ almost surely, and likewise $\liminf_n S_n = -\infty$; so $J_\lambda(X; D_n) - J_\lambda(Y; D_n) = a_X - a_Y + S_n$ changes sign infinitely often. The degenerate case is immediate. (iii) $S_n := \sum_{j \leq n}(c_X - c_Y)(z_j)$ has i.i.d. integrable increments with mean $\mu > 0$, and $J_\lambda(Y; D_n) < J_\lambda(X; D_n)$ iff $S_n > x$. So $\tau_x = \min\{n : S_n > x\}$ is a stopping time with $\mathbb{E}\tau_x < \infty$ (a random walk with positive drift), and Wald's identity gives $\mathbb{E}S_{\tau_x} = \mu\mathbb{E}\tau_x$ with $S_{\tau_x} > x$; if the increments are at most b, also $S_{\tau_x} \leq x + b$. For the path statement fix $\epsilon \in (0, \mu)$; by the strong law there is $N(\omega)$ with $|S_n - n\mu| \leq \epsilon n$ for $n \geq N$. For $x > \max_{n \leq N} S_n$ we have $\tau_x - 1 \geq N$, and $S_{\tau_x} > x \geq S_{\tau_x - 1}$ gives $x/(\mu + \epsilon) < \tau_x \leq x/(\mu - \epsilon) + 1$. Similarly $\varsigma_x \geq \tau_x - 1 \geq N$ and $S_{\varsigma_x} \leq x$ give $\varsigma_x \leq x/(\mu - \epsilon)$. Divide by x, let $x \to \infty$ and then $\epsilon \to 0$. □

The code-R cost of a silent datum is $-\log$ of a share that is uniform near 0 (the proof of Lemma A.20), so it has all moments, and (i)–(iii) cover every application in this section. The bounded-increment bound of (iii) applies to the race between T_0^{-Z} and T^{ε^*}, whose cost differences are bounded above by the largest cost of T_0^{-Z}; for the race between T^{ε^*} and T_{tr} in code R the increments are unbounded, and only the lower bound and the path limits apply.

Corollary A.23 (The phases of selection, coarse data)**.** *Take a pricing with i.i.d. per-datum costs as in Proposition A.16: exactly Q/rand and R/rand (with ρ stated), or Q/det and R/det as a constant-c_{time} cost model (Proposition A.17). Assume $\Delta \geq 2\varepsilon^*$, $\mathbb{P}[e \neq 0] > 0$, $\varepsilon^* > 0$ and $h_{\mathrm{tr}} > h$. The pool is $\{\emptyset, T_0$ (and $T_0^{-Z}), T^{\varepsilon^*}, T_{\mathrm{tr}}\}$, with $a_\emptyset = 0$, $a_{T_0} = \lambda\ell_0$, $a_{T^{\varepsilon^*}} = \lambda(\ell_0 + h)$ and $a_{T_{\mathrm{tr}}} = \lambda(\ell_0 + h_{\mathrm{tr}})$. Almost surely:*

- *(a) the exact law is refuted for the first time at a geometric index of mean $1/p_\Delta$, and until then T_0 and T_0^{-Z} coincide;*
- *(b) T_{tr} is eventually the unique λ-MAP; the first time it beats T^{ε^*} has mean above $\lambda(h_{\mathrm{tr}} - h)/m_{T^{\varepsilon^*}}$, and at most $(\lambda(h_{\mathrm{tr}} - h) + \gamma)/m_{T^{\varepsilon^*}}$ under code Q, where the costs of T^{ε^*} are bounded by γ;*
- *(c) without T_{tr}, the eventual λ-MAP is the form of least rate among $m_\emptyset$, $m_{T_0^{-}z}$ and $m_{T^{\varepsilon^*}}$ (T^{ε^*} against T_0^{-Z} by Proposition A.18, ties by Theorem A.22(ii));*
- *(d) the expected switch points are: $\emptyset \to T_0$ at $\lambda\ell_0/\gamma$ (deterministic, if T_0 is not yet refuted); $T_0 \to T_0^{-Z}$ at the first refutation, of mean $1/p_\Delta$; $T_0^{-Z} \to T^{\varepsilon^*}$, if $m_{T_0^{-}z} > m_{T^{\varepsilon^*}}$, at a time of mean above $\lambda h/(m_{T_0^{-}z} - m_{T^{\varepsilon^*}})$, and at most $(\lambda h + \gamma_{\mathrm{ex}})/(m_{T_0^{-}z} - m_{T^{\varepsilon^*}})$ when γ_{ex} is bounded; and $T^{\varepsilon^*} \to T_{\mathrm{tr}}$ as in (b).*

When these expected switch points increase in this order, the predicted order of the λ-MAPs is $\emptyset \to T_0 \to T_0^{-Z} \to T^{\varepsilon^*} \to T_{\mathrm{tr}}$.

Proof. Theorem A.22 with the costs of Proposition A.16 and Lemma A.15: refutation by T_0 has probability p_Δ per datum, independently, and while unrefuted T_0 proves every datum (it is never silent), so $J_\lambda(T_0; D_n) = \lambda\ell_0 < n\gamma = J_\lambda(\emptyset; D_n)$ once $n > \lambda\ell_0/\gamma$. T_{tr} has rate 0 and every other form a positive rate ($p_\Delta > 0$, $s_\Delta > 0$, $\gamma_{\mathrm{si}} \geq \log \mathrm{e} - 1 > 0$), which gives (b) with Theorem A.22(iii) at level $\lambda(h_{\mathrm{tr}} - h)$. The bound in (d) for $T_0^{-Z} \to T^{\varepsilon^*}$ is Theorem A.22(iii) applied to the race between these two forms from $n = 0$, at level λh. □

A simulation of the Q/det cost model ($\ell_0 = 200$, $h = 20$, $h_{\mathrm{tr}} = 2000$, $c_{\mathrm{time}} = 22$; 400 runs per setting; code in the accompanying repository) shows the five phases in this order when the predicted switch points are ordered: for a sinusoidal error at $\Delta = 60'$ and $\lambda = 1$ (rates 0.323 and 0.283) the mean switch time to the honest law is 529.8, inside the interval $[495.6, 1251.3]$ given by (d), the mean switch time to the true theory is 7010.1, inside $[6994.9, 7024.9]$, and the mean first refutation is 91.8 against 94.25; when the two rates are close, the patched and the honest law alternate several times before the honest law settles. With rates tuned to be equal, the sign of $J_\lambda(T_0^{-Z}) - J_\lambda(T^{\varepsilon^*})$ changes 148 times on average up to $n = 2 \times 10^5$, and 43% of runs still change sign after $n = 10^5$. Under SAI's pricing with exact theory-machine costs (R/rand, sinusoid, $\varepsilon^* = 1'$, $\Delta = 60'$, $\lambda = 1$; rates 0.1749 and 0.02953), the mean first time the honest law is the λ-MAP is 133.6 against a predicted switch near 138, and the mean first time the true theory is is 67060 against 67055; all 400 runs reach it.

Proposition A.24 (Humility)**.** *Let* $\Pi_\lambda(T \mid D) \propto 2^{-J_\lambda(T;D)}$ *be the power-prior posterior (Definition 3.6 of the companion paper on steeper priors) restricted to a finite pool of deterministic forms (code Q) containing* $\emptyset$, T_0^{-Z}, T^{ε^*} *and possibly* T_{tr}*, and ask a new question at a fresh time* t'*, with cell width* $\Delta' \leq 2\varepsilon^*$.

(a) The pool's three-valued prediction (the analogue, for the pool, of the trichotomy of **??***) puts mass* $\Pi_\lambda(T^{\varepsilon^*} \mid D) + \Pi_\lambda(\emptyset \mid D)$ *on "independent", mass* $\Pi_\lambda(T_0^{-Z} \mid D)$ *on the cell of* $f(t')$*, which is wrong with probability* $p_{\Delta'}$ *over the dither, and mass* $\Pi_\lambda(T_{\mathrm{tr}} \mid D)$ *on the true cell.*

(b) Without T_{tr}*, if the least rate at the training cell width* Δ *is attained by one form only, the posterior concentrates on it almost surely: the inducer ends humble (independent mass tending to* 1 *on finer questions) if* T^{ε^*} *or* $\emptyset$ *is selected, and overconfident (a wrong cell with probability* $p_{\Delta'}$*) if* T_0^{-Z} *is; Proposition A.18 at* Δ *and Proposition A.21 decide which. If two forms tie in rate with a non-degenerate cost difference of finite variance, the posterior oscillates between them forever.*

(c) In a randomised pool (code R) a band-confined completion $\mathrm{BC}_{\mathcal{G}}$ *is not silent: it puts on each cell its* $\mathcal{G}$*-mass, a spread over the cells its band meets and never probability one on a wrong cell; the exact law with random exceptions puts probability* $1 - \rho$ *on the cell of* $f(t')$.

Proof. (a) T^{ε^*} is silent on every question with $\Delta' \leq 2\varepsilon^*$ (Lemma A.15), and $\emptyset$ is silent. T_0^{-Z} proves the cell of $f(t')$, since t' is fresh and so not a patched time, and T_{tr} proves the true cell. (b) Theorem A.22(i), (ii), with $\Pi_\lambda(X \mid D)/\Pi_\lambda(Y \mid D) = 2^{-(J_\lambda(X;D) - J_\lambda(Y;D))}$, whose exponent diverges linearly when the rates differ. (c) The definition of $\mathrm{BC}_{\mathcal{G}}$, and the withdrawal probability ρ. □

body	ε^*, minimax fit (12 / 100 / 2000 yr)	pk, minimax fit	pk, mean fit
Mercury	$0.38'$ / $2.02'$ / $37.4'$	2.80–3.19	3.29–4.71
Venus	$0.36'$ / $0.47'$ / $3.6'$	2.27–3.65	2.81–4.07
Earth–Moon barycentre	$0.27'$ / $0.62'$ / $7.3'$	2.34–3.95	2.91–4.03
Mars	$1.44'$ / $3.04'$ / $49.9'$	2.71–3.38	3.33–4.47
Jupiter	$0.70'$ / $6.57'$ / $34.0'$	1.60–2.65	3.00–7.08
Saturn	$0.17'$ / $7.46'$ / $81.9'$	1.66–2.39	3.21–6.91
Moon (1 month / 1 yr / 18.6 yr)	$30.9'$ / $156'$ / $480'$	1.89–2.21	1.89–3.17

Table 1: Longitude errors of one fixed Kepler ellipse per body and window, against the plan94 and moon98 ephemerides (not the sky): the largest error of the minimax fit, and the peakedness of each fit over the windows of 12 to 2000 years (for the Moon, one month to 18.6 years). Largest longitude errors of the minimax fit; the least-squares fits of **??** measure angular separation and differ. Computed; code in the accompanying repository.

The Kepler residuals. Table 1 summarises the error laws. For each body and window, one fixed Kepler ellipse (seven elements) was fitted four ways (three-dimensional least squares, least squares on longitude and latitude, a robust fit for small mean error, and an L_p continuation for small maximum error), and two series of longitude errors at 8000 uniform times were kept: that of the fit with the least sample maximum (the *minimax fit*) and that of the fit with the least sample mean (the *mean fit*). The ephemerides are plan94 for the planets and moon98 for the Moon. An honest law whose tolerance is the sample maximum is slightly false if the window's true supremum is larger. A local parabola through the three time-sorted samples around the maximum estimates the excess at a median 2×10^{-4} (relative) and at most 1.9% in 33 of the 39 series apart from one, Mercury over 2000 years, where the estimate is unreliable because the sampling interval of about 91 days is comparable to Mercury's 88-day period. The band machines therefore use $\varepsilon = (1 + \kappa_\mathrm{b})\varepsilon^*$ with $\kappa_\mathrm{b} \in \{0, 0.01, 0.05\}$, and every comparison below comes out the same for the three values. The rates are computed at eleven cell widths from $0.1'$ to $30°$, so that there are 429 (body, window, cell width) cells. Within the pool $\{\emptyset, T_0^{-Z}, T^{\varepsilon^*}\}$ under R/rand (and under R/det at $\lambda = 1$ and 4), the uniform band has the least rate in all 429 cells, with exact closed-form costs; in that pool's tolerance family with whole-circle exceptions the optimum is a tolerance of about 0.7–1.0 ε^* with at most 1% of the data as exceptions. With band-confined completions, the histogram machine SH has the least rate in all 429 cells at each κ_b, between 0.4% and 25.7% below the uniform band (median 6.6%), and lies within 0.0011 bits of the Gibbs bound of its test half in every coarse-regime pair; the best two-point form beats the uniform band in 379 of the 429 cells at $\kappa_\mathrm{b} = 0.01$, and in the coarse regime in 34 of the 39 body–window pairs, with best ρ between 0.52 and 0.997, exactly where the criterion of Proposition 1.2(d) says. The five pairs in which the uniform band beats every two-point form are Jupiter over 12 and 30 years, Saturn over 100 years and the Moon over one month and 18.6 years, whose minimax-fit peakedness is 1.60–1.96. The gain persists at fine precision: for the Earth–Moon barycentre over 100 years at $\Delta = 1'$ the rates of the uniform band, the best two-point form and SH are 0.651, 0.560 and 0.503 bits per datum, and for Mars over 12 years 1.660, 1.558 and 1.496. Under Q/det (naming a time costs $c_\mathrm{time} = \log(\text{days in the window}) + 2$ bits, 14.1–21.5 for the planets and 6.8, 10.5 and 14.7 for the Moon's windows; a finite-horizon cost model) the threshold $(\lambda c_\mathrm{time} + \gamma)/(2\gamma)$ grows as the cells coarsen. At $\lambda = 1$ the exact law with exceptions wins from cell widths of about ε^* upwards, the honest law only at the coarsest cells (30°, and from 10° for Jupiter over 30 years and Saturn over 2000 years) and not even there for 17 of the 39 pairs, and at finer cells no law pays; the best

member of the tolerance family is then a small tolerance (0.05–0.5 ε^*) with exception shares up to tens of per cent. At $\lambda = 2$ the honest law takes over at cell widths from $8.1'$ to $962'$ (median $134'$), and at $\lambda = 4$ from $0.72'$ to $962'$ (median $13.4'$). Under Q/rand the exact law with random exceptions wins almost everywhere. The residuals are deterministic functions of time, so a better theory (Newton's, or the ephemeris itself) eventually beats every Kepler-based machine.

A.5 Derivation horizons and sound reasoning with approximate models

Work over $B = \mathrm{RCF}$ with constants $x_0, x_1, \ldots$ for the states of a system in $\mathbb{R}^m$ (one constant per coordinate), a rational initial state a, a map Ψ definable in B, and a norm $|\cdot|$ for which B proves that Ψ is L-Lipschitz (for a linear $\Psi(x) = Ax$ with rational A and the sup-norm, $L = \|A\|_\infty$, provably in RCF). The *honest dynamical law* is $T^\varepsilon_\Psi := \{|x_0 - a| \le \varepsilon^{(0)}\} \cup \{|x_{s+1} - \Psi(x_s)| \le \varepsilon : s \ge 0\}$, and the *exact law* is $T^0_\Psi := \{x_0 = a\} \cup \{x_{s+1} = \Psi(x_s) : s \ge 0\}$.

Proposition A.25 (Derivation horizon)**.**

(a) For every $L \ge 0$, $T^\varepsilon_\Psi \vdash |x_r - \Psi^r(a)| \le \varepsilon^{(r)}$, with $\varepsilon^{(r)} := L^r\varepsilon^{(0)} + \varepsilon(L^r - 1)/(L - 1)$ for $L \ne 1$ and $\varepsilon^{(r)} := \varepsilon^{(0)} + r\varepsilon$ for $L = 1$.

(b) Let $\Psi(x) = Ax$ be linear, with sup-norm tolerances. The values of the i-th coordinate $x_{r,i}$ over the models of $B \cup T^\varepsilon_\Psi$ form exactly the interval $(A^r a)_i \pm h_{r,i}$, where $h_{r,i} := \varepsilon^{(0)}\|(A^r)_{i\cdot}\|_1 + \varepsilon \sum_{s<r}\|(A^s)_{i\cdot}\|_1$ (the projection of a zonotope). So T^ε_Ψ proves "$x_{r,i} \in [v - \delta, v + \delta]$" iff that interval contains $(A^r a)_i \pm h_{r,i}$. In dimension one, $h_r = \varepsilon^{(r)}$ with $L = |A|$, so (a) is tight.

(c) Put $r^*(\delta) := \max\{r : \varepsilon^{(r)} \le \delta\}$. For $\varepsilon^{(0)} = 0$: $r^*(\delta) = \lfloor \delta/\varepsilon \rfloor$ for $L = 1$; $r^*(\delta) = \lfloor \log_L(1 + \delta(L - 1)/\varepsilon) \rfloor$ for $L > 1$; and for $L < 1$, $r^*(\delta) = \infty$ iff $\delta \ge \varepsilon/(1 - L)$, and $r^*(\delta) = \lfloor \log_L(1 - \delta(1 - L)/\varepsilon) \rfloor$ otherwise. For every L-Lipschitz Ψ, T^ε_Ψ proves a datum at precision δ about x_r, centred at $\Psi^r(a)$, if $r \le r^*(\delta)$; for linear Ψ in dimension one with $L = |A|$, also only if.

Proof. (a) By induction, $|x_{s+1} - \Psi^{s+1}(a)| \le |x_{s+1} - \Psi(x_s)| + |\Psi(x_s) - \Psi(\Psi^s(a))| \le \varepsilon + L\varepsilon^{(s)}$; each step is an instance of the axioms and of the provable Lipschitz inequality, and the recursion $\varepsilon^{(s+1)} = L\varepsilon^{(s)} + \varepsilon$ has the stated solution. (b) $x_r - A^r a = A^r(x_0 - a) + \sum_{s<r} A^{r-1-s}u_s$ with $u_s := x_{s+1} - Ax_s$, and in the real model the deviations $x_0 - a$ and u_s range freely over sup-norm boxes of half-widths $\varepsilon^{(0)}$ and ε. A linear image of a product of boxes is a zonotope, which is convex; its projection on coordinate i is the interval of half-width the sum of the ℓ_1-norms of the corresponding rows, with the extreme values attained by sign choices. So exactly the stated interval of values occurs in $\mathbb{R}$. Since RCF is complete and the claim about $x_{r,i}$ involves only the finitely many axioms with $s < r$, T^ε_Ψ proves an interval statement about $x_{r,i}$ iff it holds in every real model, that is iff the interval contains $(A^r a)_i \pm h_{r,i}$. (c) Solve $\varepsilon^{(r)} \le \delta$; the "if" is (a), and the "only if" for linear Ψ in dimension one is (b). □

For non-linear Ψ the "only if" fails. Let $\Psi(x) = 0$ on $[-1, 1]$ and $\Psi(x) = 2(x - \operatorname{sign} x)$ outside, so that $L = 2$, with $a = 0$, $\varepsilon^{(0)} = 0$ and $\varepsilon = 1/10$. Then $|x_s| \le 1/10 \le 1$ gives $\Psi(x_s) = 0$ and so $|x_{s+1}| \le 1/10$; by induction T^ε_Ψ proves "$x_r \in [-1/10, 1/10]$" at every depth, while $\varepsilon^{(r)} = (2^r - 1)/10$ gives $r^*(1/10) = 1$. A computation (code in the accompanying repository) confirms the closed form against exact interval propagation for $r \le 200$ and several L between 0.5 and 2; at $\delta/\varepsilon = 10^4$ the horizons are 10^4, 463, 72 and 13 for $L = 1$, 1.01, 1.1 and 2, and infinite for $L \le 0.99$. In dimensions 3 and 10, for random matrices of spectral radius 0.95–1.05, propagating boxes by interval arithmetic overestimates the exact tolerance $h_{r,i}$ by a median factor of 5–300 at $r = 10$

and up to about 10^{10} at $r = 30$, and the ∞-norm Lipschitz bound by up to about 7×10^{13}: a reasoner that propagates boxes has a much shorter effective horizon than the theory.

Remark A.26 (Typical against worst case). With $L = 1$ and $r = 100$ steps the propagated tolerance is 100ε, while with independent random-sign errors $\pm\varepsilon$ the mean deviation is $\varepsilon r\binom{r}{r/2}2^{-r} \approx 7.96\varepsilon$: a probabilistic error model answers much further out, with a stated confidence instead of a guarantee.

Proposition A.27 (The horizon of an exact approximation)**.** *Suppose the true trajectory satisfies T^{ε}_{Ψ}, but the inducer holds the exact law T^{0}_{Ψ} with the same a. Then T^{0}_{Ψ} is consistent and proves $x_r = \Psi^r(a)$, which is within $\varepsilon^{(r)}$ of the true x_r. If $r \le r^*(\delta)$, the depth-r conclusions of T^{0}_{Ψ} are δ-correct for every trajectory compatible with the honest law. For linear Ψ in dimension one with $L = |A|$, some trajectory compatible with the honest law attains the deviation $\varepsilon^{(r)}$, so in that case they are δ-correct for all such trajectories only if $r \le r^*(\delta)$. And T^{0}_{Ψ} together with one true datum about x_r whose interval excludes $\Psi^r(a)$ is inconsistent, and so proves every sentence.*

Proof. The exact trajectory is a model of T^{0}_{Ψ}, and the true trajectory is a model of T^{ε}_{Ψ}. The middle claims are Proposition A.25(a), (b). For the last, the datum and $x_r = \Psi^r(a)$ are jointly refutable in B. □

Proposition A.28 (Sound reasoning with approximate models)**.** *Let $T_i := \{\forall t \in \mathrm{Dom}_i : |q(t) - f_i(t)| \le \varepsilon_i\}$, $i \le k$, be honest pointwise models of one quantity q (scalar, or vector-valued with a norm), with q otherwise unconstrained, and let T^{0}_{i} be their exact versions ($\varepsilon_i = 0$).*

(a) Let B be true in the intended structure. If every T_i is true, then $\bigcup_i T_i$ is consistent and every sentence derivable from $B \cup \bigcup_i T_i$ is true; every derived statement about $q(t)$ is a set containing the true value.

For (b)–(d) assume moreover that B decides every sentence built from the f_i, the Dom_i and the ε_i at the times concerned (for example $B = \mathrm{RCF}$, with semialgebraic f_i and Dom_i and rational ε_i).

(b) $\bigcup_i T^{0}_{i}$ is inconsistent, and so proves every sentence, iff $f_i(t) \ne f_j(t)$ for some $t \in \mathrm{Dom}_i \cap \mathrm{Dom}_j$.

(c) $\bigcup_i T_i$ is consistent iff for each t the balls $\{v : |v - f_i(t)| \le \varepsilon_i\}$ with $t \in \mathrm{Dom}_i$ have a common point. For scalar q, and for vector q with sup-norm (box) tolerances, this holds iff it holds pairwise, $|f_i(t) - f_j(t)| \le \varepsilon_i + \varepsilon_j$ on every overlap. For Euclidean tolerances in $\mathbb{R}^m$, $(m+1)$-wise intersection is necessary and sufficient, and pairwise intersection is not sufficient.

(d) If B proves that a map Ψ is L_Ψ-Lipschitz, then $T_i \vdash |\Psi(q(t)) - \Psi(f_i(t))| \le L_\Psi\varepsilon_i$.

Proof. (a) The intended structure is a model, and derivations preserve truth. (b) If two exact models disagree at a common time t, together they prove $f_i(t) = f_j(t)$, which B refutes by the hypothesis. If they agree on all overlaps, then in every model of B (in which B's decisions fix the values $f_i(t)$) setting $q(t) := f_i(t)$ for $t \in \mathrm{Dom}_i$, and arbitrarily elsewhere, gives a model. (c) The constants $q(t)$ for distinct t are distinct and otherwise unconstrained, so by compactness $B \cup \bigcup_i T_i$ is consistent iff for each t, B is consistent with "some v lies in all the balls at t". By the hypothesis B decides this sentence, and B is true in the intended structure, so it is consistent with B iff it holds there, that is iff the balls have a common point. Intervals that meet pairwise have $\max_i$(left end) $\le \min_j$(right end), which is a common point (Helly's theorem in dimension one), and boxes are products of intervals. Helly's theorem in $\mathbb{R}^m$ (standard; Helly, 1923) gives the

$(m+1)$-wise criterion. Three discs of radius 1.1 centred at the vertices of an equilateral triangle of side 2 meet pairwise but have no common point, since the circumradius is $2/\sqrt{3} \approx 1.155$. (d) is the Lipschitz inequality in B. □

Without the hypothesis on B, (b) and (c) fail: take $B = \emptyset$, $f_i(t)$ and $f_j(t)$ distinct constant symbols and $\mathrm{Dom}_i = \mathrm{Dom}_j$ a single time; then $T_i^0 \cup T_j^0$ is consistent (interpret the two constants as equal), although the values differ in the intended structure. A computation finds pairwise and joint overlap in agreement on 10^5 random families of 2 to 6 intervals and on 2×10^4 random families of boxes in $\mathbb{R}^2$.

A.6 Best systems

Definition A.29 (Explanatory power)**.** For a theory T and a module $\mathcal{A} \subseteq T$, $\mathrm{EP}_D(\mathcal{A}; T) := \big(\mathrm{Exc}_D(T \setminus \mathcal{A}) - \mathrm{Exc}_D(T)\big)/\|\mathcal{A}\|$: the bits of data the module explains per bit of its statement, in the context T.

Proposition A.30 (Marginal slopes)**.** *Let T^* be a λ-MAP within a family of theories that contains $T^* \setminus \mathcal{A}$ and $T^* \cup \mathcal{B}$ for the modules considered. For every module $\mathcal{A} \subseteq T^*$, $\mathrm{Exc}_D(T^* \setminus \mathcal{A}) - \mathrm{Exc}_D(T^*) \geq \lambda\|\mathcal{A}\|$; and for every $\mathcal{B}$ disjoint from T^*, $\mathrm{Exc}_D(T^*) - \mathrm{Exc}_D(T^* \cup \mathcal{B}) \leq \lambda\|\mathcal{B}\|$.*

Proof. $J_\lambda(T^*; D) \leq J_\lambda(T^* \setminus \mathcal{A}; D)$ and $J_\lambda(T^*; D) \leq J_\lambda(T^* \cup \mathcal{B}; D)$, with $\|T^*\| - \|T^* \setminus \mathcal{A}\| = \|\mathcal{A}\|$ and $\|T^* \cup \mathcal{B}\| - \|T^*\| = \|\mathcal{B}\|$. □

Proof of Proposition 1.3. By the hypotheses (Proposition A.8) the hull lemma (Lemma 3.8(v) of the companion paper on steeper priors) applies: for $\lambda \in (\sigma_{i+1}, \sigma_i)$ the λ-MAPs are exactly the theories at v_i, and at $\lambda = \sigma_j$ $(j \geq 1)$ they are the theories on the closed edge $[v_{j-1}, v_j]$. (b)⇒(a): $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$. (a)⇒(b): if some edge slope σ_j lay in (τ, σ), the λ-MAP set at $\lambda = \sigma_j$, the theories on $[v_{j-1}, v_j]$, would differ from the set just below σ_j, the theories at v_j only, since $v_{j-1} \neq v_j$. So no edge slope lies in (τ, σ), and (τ, σ) lies in one interval (σ_{i+1}, σ_i), which is (b). (b)⇔(c) is Corollary 5.4(c) of the companion paper on steeper priors: a zero-window (σ, τ)-kink with $\tau < \sigma$ is a vertex v_i with $(\tau, \sigma) \subseteq (\sigma_{i+1}, \sigma_i)$, and conversely every vertex v_i is a zero-window (σ_i, σ_{i+1})-kink, hence a (σ, τ)-kink whenever $\sigma \leq \sigma_i$ and $\tau \geq \sigma_{i+1}$. □

A computation on 400 random point families with 10 random intervals each, testing "the λ-MAP set is constant on (τ, σ)" exactly at every crossing point and between consecutive ones, agrees with (b) in all 4000 cases.

Proposition A.31 (The trivialisation is blocked)**.** *Let $\mathcal{L}' \supseteq \mathcal{L}$ be any extension of the language, for example by a predicate* Act *true of exactly the actual things, with the axiom $\forall x\, \mathrm{Act}(x)$; fix a syntax code for $\mathcal{L}'$, and let the data be base-language answers whose provability from the theories considered is decidable. Then every finite theory T in $\mathcal{L}'$ consistent with D has $\|T\| + \mathrm{Exc}_D^Q(T) \geq K(y_D \mid \mathrm{dom}\, D) - K(\mathcal{L}') - c$, where $K(\mathcal{L}')$ is the complexity of a decoder for $\mathcal{L}'$ and its syntax code. Without decidability, add n on the left. If $\mathcal{L}'$ and its code are fixed independently of D, $K(\mathcal{L}')$ is a constant.*

Proof. The weights in the proof of Proposition A.9 search for, or decide, proofs from T of base-language answers. Given a decoder for the syntax of $\mathcal{L}'$ they are computable (respectively lower semicomputable) from T and the questions, and they do not otherwise depend on the language. So the argument of Proposition A.9(b) (respectively (a)) gives $K(y_D \mid \mathrm{dom}\, D) \leq K(\mathcal{L}') + K(T \mid \mathcal{L}') + \mathrm{Exc}_D^Q(T) + c$ (respectively with n added), and $K(T \mid \mathcal{L}') \leq \|T\| + c$. □

Proposition A.32 (Facts used once never enter for $\lambda > 1$)**.**

(a) Let T^* be a λ-MAP with $\lambda > 1$ and χ a sentence. Suppose that, in the context $T^* \setminus \{\chi\}$, adding χ turns exactly the exceptions in a set $\mathcal{D}_\chi \subseteq D$ into proved data and changes nothing else, and that $\sum_{(\mathcal{Q},a)\in\mathcal{D}_\chi} \gamma(\mathcal{Q}) < \lambda\ell(\chi)$. Then $\chi \notin T^*$. In particular this applies to a datum's answer a stated as an axiom whenever $\gamma(\mathcal{Q}) \le \ell(a)$ and stating it proves no other datum in that context.

*(b) Let provability on the data be decidable in the class, with κ the constant of Proposition A.9, and let T_s be b-*sufficient*: $\|T_s\| + \mathrm{Exc}_D(T_s) \le K(y_D \mid \mathrm{dom}\, D) + b$. Every theory T has $\mathrm{Exc}_D(T) \ge \mathrm{Exc}_D(T_s) - (\|T\| - \|T_s\|) - (b+\kappa)$; a module $\mathcal{A}$ added to T_s has $\mathrm{EP}_D(\mathcal{A}; T_s \cup \mathcal{A}) \le 1 + (b+\kappa)/\|\mathcal{A}\|$; and every λ-MAP T with $\lambda > 1$ has $\|T\| \le \|T_s\| + (b+\kappa)/(\lambda-1)$.*

Proof. (a) If $\chi \in T^*$, deleting it raises Exc_D by exactly $\sum_{\mathcal{D}_\chi} \gamma(\mathcal{Q}) < \lambda\ell(\chi) = \lambda\|\{\chi\}\|$, contradicting Proposition A.30. For an answer stated as an axiom, $\mathcal{D}_\chi$ is the one datum, and $\gamma(\mathcal{Q}) \le \ell(a) < \lambda\ell(a)$. (b) Proposition A.9(b) gives $\|T\| + \mathrm{Exc}_D(T) \ge K(y_D \mid \mathrm{dom}\, D) - \kappa \ge \|T_s\| + \mathrm{Exc}_D(T_s) - b - \kappa$, which is the first claim, a right line of slope 1 through the point of T_s. For a module $\mathcal{A}$ disjoint from T_s, $\|T_s \cup \mathcal{A}\| - \|T_s\| = \|\mathcal{A}\|$, which gives the second. For a λ-MAP T, $\lambda\|T\| + \mathrm{Exc}_D(T) \le \lambda\|T_s\| + \mathrm{Exc}_D(T_s)$; combined with the first claim this gives $(\lambda-1)(\|T\| - \|T_s\|) \le b + \kappa$, which is Corollary 5.5(a) of the companion paper on steeper priors in this class. □

So at $\lambda = 1$ a fact that explains only itself is excluded only by the code's per-axiom overhead, while every $\lambda > 1$ excludes every module whose explanatory power is below λ.

The solar-system toy. Bodies $b \le k_\mathrm{b}$ have n_b observations each, each of literal cost γ (code Q), and $n_\mathrm{tot} := \sum_b n_b$. The modules are the law Λ (L bits), body b's elements El_b (k bits), and Kepler's third law K3 (L_3 bits), which shortens every elements module by s bits, $0 < s < k$, because it derives the period from the semi-major axis. $\Lambda \cup \mathrm{El}_b$ proves all of b's observations, and no other combination proves any observation: laws alone answer no position question, and neither do elements alone. The theories are Λ or not, K3 or not, and a set S of bodies whose elements are axioms.

Proposition A.33 (Laws and initial conditions in the solar-system toy)**.** *Put $k_\Lambda := k$, $k_{\Lambda+\mathrm{K3}} := k - s$ and $S_x(\lambda) := \{b : \gamma n_b > \lambda k_x\}$ for $x \in \{\Lambda, \Lambda + \mathrm{K3}\}$. The optimal values of the three law choices are*

$$J^*_\emptyset = \gamma n_\mathrm{tot}, \qquad J^*_\Lambda = \lambda L + \sum_b \min(\gamma n_b, \lambda k), \qquad J^*_{\Lambda+\mathrm{K3}} = \lambda(L + L_3) + \sum_b \min(\gamma n_b, \lambda(k-s)),$$

*attained by $S = S_x(\lambda)$ (bodies with $\gamma n_b = \lambda k_x$ may go either way); the λ-MAP is the best of the three (*K3 *without Λ is useless).*

(a) Λ beats the empty theory iff $\sum_{b\in S_\Lambda(\lambda)}(\gamma n_b - \lambda k) > \lambda L$.

(b) $J^*_{\Lambda+\mathrm{K3}} - J^*_\Lambda = \lambda L_3 - \sum_b \big[\min(\gamma n_b, \lambda k) - \min(\gamma n_b, \lambda(k-s))\big]$. If no body has $\lambda(k-s) < \gamma n_b \le \lambda k$, so that S is the same with and without K3*, then $J^*_{\Lambda+\mathrm{K3}} < J^*_\Lambda$ iff $s|S| > L_3$, whatever λ is.* K3 *is in the λ-MAP iff in addition $J^*_{\Lambda+\mathrm{K3}} < J^*_\emptyset$.*

(c) In the λ-MAP $\Lambda \cup \{\mathrm{El}_b : b \in S\}$ (with or without K3*), $\mathrm{EP}(\Lambda) = \gamma \sum_{b\in S} n_b/L$ and $\mathrm{EP}(\mathrm{El}_b) = \gamma n_b/k_x$.*

Proof. Without Λ no observation is proved, so the best such theory is empty, with $J = \gamma n_{\text{tot}}$. For a law choice x with Λ, J_λ is a constant plus a sum over bodies, where including El_b costs λk_x and excluding it costs γn_b; minimise each term. (a) is the comparison of J^*_Λ with $J^*_\emptyset$. (b) The bracket is λs when $\gamma n_b > \lambda k$, $\gamma n_b - \lambda(k-s)$ when $\lambda(k-s) < \gamma n_b \le \lambda k$, and 0 otherwise; without bodies in the middle range the difference is $\lambda(L_3 - s|S|)$. The λ-MAP is the best of the three law choices, the empty one included. (c) Deleting Λ (respectively El_b) makes exactly the observations of the bodies in S (respectively of b) literal. □

Part (b) compares law choices; it is not membership of the λ-MAP. With two bodies of one observation each, $\gamma = 10$, $L = 1000$, $k = 5$, $s = 4$, $L_3 = 1$ and $\lambda = 1$, S is both bodies with and without K3 and $s|S| = 8 > L_3$, but the λ-MAP is the empty theory: $J^*_\emptyset = 20$ against $J^*_{\Lambda+\text{K3}} = 1003$ and $J^*_\Lambda = 1010$. The middle-range condition of (b) matters too: in a population of eight planets with 655 to 3838 observations, sixty asteroids with a median of about five and thirty comets with one or two ($L = 300$, $L_3 = 150$, $k = 60$, $s = 15$, $\gamma = 12.4$), K3 is in the λ-MAP at $\lambda = 6.75$ although $s|S_\Lambda(\lambda)| = 135 < L_3$, because with K3 more bodies pay for their elements. A computation (code in the accompanying repository) confirms the closed form against brute force over all theories in 2100 random cases, and the criterion of (b) in all 13250 random cases with the same S under both law choices.

In the same population the elements of a well-observed body explain hundreds of bits per bit (Proposition A.33(c)) and sit on the steep side of the hull with the law, and a computation (code in the accompanying repository) finds that the λ-MAP is "the law plus the elements of all eight planets" for every $\lambda \in (7.03, 135.37)$, while below $\lambda \approx 7$ it changes with every asteroid. Inside such an interval the worry that the best system depends on an arbitrary choice of balance (cf. Woodward, 2014) is moot, and robustness to the balance (Proposition 1.3) is a checkable property of a language, a family of candidates and a data set.

Proposition A.34 (Laws and initial conditions in the question format)**.** *Let the data be conditional questions "given the elements α_b of body b, is ℓ the case?", with compatibility as in* (E3)*: a datum is compatible with T when $B \cup T \cup \{\alpha_b\}$ is consistent and proves ℓ. In the toy, every theory of the family (the law,* K3*, and elements modules stating the true elements) is consistent with every true antecedent. Then an elements module El_b with the same content as α_b adds nothing, so $\text{EP}(\text{El}_b) = 0$ and no λ-MAP contains elements; and Λ alone answers every question, with $\text{EP}(\Lambda) = \gamma n_{\text{tot}}/L$ however the n_{tot} observations are spread over bodies.*

Proof. Immediate from the definition of compatibility: $T \cup \{\alpha_b\} \cup \text{El}_b$ and $T \cup \{\alpha_b\}$ prove the same sentences, and $\Lambda \cup \{\alpha_b\}$ proves all of b's observations. Adding El_b raises J_λ by $\lambda k > 0$ and lowers no exception cost. □

Without the consistency condition the encoding is degenerate: a theory that refutes the antecedents would prove every datum (**??**). In the question format the distinction between laws and initial conditions is the distinction between theory and question antecedent.

A.7 Explanatory power, unification and correction

Proposition A.35 (Explanatory power as a likelihood ratio; unification)**.** *Write $P_T(D) := 2^{-\text{Exc}_D(T)}$ (code Q or R). For a hypothesis h, a background theory $T_\flat$, data sets D_1, D_2 on disjoint questions and a theory x, put $I(h; D \mid T_\flat) := \log(P_{T_\flat \cup h}(D)/P_{T_\flat}(D))$, $I_x(D_1; D_2) := \log \frac{P_x(D_1 \cup D_2)}{P_x(D_1)P_x(D_2)}$ and $U(D_1, D_2; h \mid T_\flat) := I(h; D_1 \cup D_2 \mid T_\flat) - I(h; D_1 \mid T_\flat) - I(h; D_2 \mid T_\flat)$.*

(a) $\|\mathcal{A}\| \, \text{EP}_D(\mathcal{A}; T) = I(\mathcal{A}; D \mid T \setminus \mathcal{A})$.

(b) $U(D_1, D_2; h \mid T_\flat) = I_{T_\flat \cup h}(D_1; D_2) - I_{T_\flat}(D_1; D_2)$. *If P is a product over questions for both theories, then $U = 0$ for every split.*

(c) *Let h be a law family with a k-bit parameter θ under the uniform prior, $P_h(D) := 2^{-k} \sum_\theta P_{h \cup \{\theta\}}(D)$, and $P_\emptyset$ a product. If each of D_1, D_2 is proved by $h \cup \{\theta^*\}$ and has likelihood 0 under every other θ, then $U(D_1, D_2; h) = k$. If $\theta = (\theta_1, \theta_2)$ with D_i depending only on θ_i, then $U = 0$. If $\theta_i = (\theta_{\mathrm{sh}}, \theta_i')$ share a k_{sh}-bit constant θ_{sh} and each D_i identifies $(\theta_{\mathrm{sh}}, \theta_i')$, then $U = k_{\mathrm{sh}}$.*

Proof. (a) is the definition of EP. (b) Expand the six logarithms: the first term of the difference is $\log P_{T_\flat \cup h}(D_1 \cup D_2) - \log P_{T_\flat \cup h}(D_1) - \log P_{T_\flat \cup h}(D_2)$, and the second is the same expression for $T_\flat$; a product law makes $I_x(D_1; D_2) = 0$. (c) Use (b) with $I_\emptyset = 0$. In the first case $P_h(D_1) = P_h(D_2) = P_h(D_1 \cup D_2) = 2^{-k}$, so $I_h(D_1; D_2) = k$. In the second, $P_h(D_1 \cup D_2) = P_h(D_1) P_h(D_2)$. In the third, with θ_i' of k' bits, $P_h(D_i) = 2^{-k_{\mathrm{sh}} - k'}$ and $P_h(D_1 \cup D_2) = 2^{-k_{\mathrm{sh}} - 2k'}$, so $I_h(D_1; D_2) = k_{\mathrm{sh}}$. □

A computation with $k = 6$ and observations given by a random table gives $U = 0.00$ for a stated parameter, 6.00 for a marginalised parameter and one body, 0.00 for two bodies with independent parameters, and 3.00 for two bodies sharing a 3-bit constant; the identity (b) holds to 3×10^{-15} on 1000 random joint laws. Myrvold describes unification as the degree to which a hypothesis renders phenomena informationally relevant to one another, a contribution to evidential support (Myrvold, 2003); by (b), U is the change in the mutual information between D_1 and D_2 that adding h brings about, which has that form. We do not claim that his formula coincides with U in every detail. At $\lambda = 1$ a module pays iff its explanatory power exceeds 1, which is the compression criterion of inductive logic programming (cf. Muggleton, 1995; Srinivasan et al., 1994).

Copernicus against Ptolemy (for the history, see Kuhn, 1957) is unification without better prediction. The lengths in the following comparison are not measured.

Proposition A.36 (Unification without better prediction)**.** *Consider three theories with a common part:* Pt_f*, Ptolemy's system with free epicycle phases, which pays for the coincidence of each superior planet p's retrograde motion with its opposition in its fit, a_p bits per planet;* Pt_s*, the same with the coincidences stated as axioms, $\sum_p a_p$ law bits more than* Pt_f *and with the fit of* Cop*; and* Cop*, Copernicus's system, H law bits more than* Pt_f *and with the fit of* Pt_s*.*

(a) Cop *beats* Pt_s*, which makes the same predictions, at every λ iff $H < \sum_p a_p$.*

(b) Cop *is* Pt_f *plus a module with* $\mathrm{EP} = \sum_p a_p / H$*, so it beats* Pt_f *iff $\lambda < \sum_p a_p / H$.*

(c) Pt_s *beats* Pt_f *iff $\lambda < 1$.*

Hence if $H < \sum_p a_p$, Cop *is the best of the three exactly for $\lambda < \sum_p a_p / H$, and* Pt_f *above; if $H \geq \sum_p a_p$,* Cop *is never strictly best:* Pt_s *is best for $\lambda < 1$ (tying* Cop *if $H = \sum_p a_p$) and* Pt_f *for $\lambda > 1$.*

Proof. Measure lengths and fits relative to Pt_f and write $a := \sum_p a_p$. Then $J_\lambda(\mathrm{Pt}_f) = 0$, $J_\lambda(\mathrm{Pt}_s) = \lambda a - a$ and $J_\lambda(\mathrm{Cop}) = \lambda H - a$. (a) $J_\lambda(\mathrm{Cop}) - J_\lambda(\mathrm{Pt}_s) = \lambda(H - a)$, negative for every $\lambda > 0$ iff $H < a$. (b) $J_\lambda(\mathrm{Cop}) - J_\lambda(\mathrm{Pt}_f) = \lambda H - a$, and the module of H bits lowers the fit by a. (c) $J_\lambda(\mathrm{Pt}_s) - J_\lambda(\mathrm{Pt}_f) = (\lambda - 1)a$. If $H < a$, then $a/H > 1$: for $\lambda < a/H$, Cop beats both others; for $\lambda > a/H > 1$, Pt_f beats both. If $H \geq a$: for $\lambda < 1$, Pt_s beats Pt_f and is at least as good as Cop, with equality iff $H = a$; for $\lambda > 1 \geq a/H$, Pt_f beats both; at $\lambda = 1$, Pt_s and Pt_f tie and Cop is no better. □

Newton's theory does not entail Kepler's laws; with the planets' masses and initial conditions it contradicts their exact form (cf. Duhem, 1954) and derives corrected statements, and Kepler's

laws within tolerances on windows, as in the model of reduction to a corrected reduced theory (cf. Schaffner, 1967); deriving "Kepler within ε on a window" from Newton's theory itself needs validated perturbation bounds (**??**). The following toy makes the structure provable. Over $B = \mathrm{RCF}$ with constants $x_{t,1}, x_{t,2}$ for $t \le \bar{t}$ (a planar state x_t per step), let Rot be the rotation with cosine $3/5$ and sine $4/5$, $\mathbf{P}$ a rational 2×2 matrix with operator norm $\|\mathbf{P}\|$, $\eta_c := 1/100$, $A := \mathrm{Rot} + \eta_c \mathbf{P}$ and $a := (1, 0)$; $|\cdot|$ is the Euclidean norm, and $|v| \le \varepsilon$ is stated as $|v|^2 \le \varepsilon^2$ with ε^2 rational. The *old exact law* is $T_0 := \{x_t = \mathrm{Rot}^t a : t \le \bar{t}\}$ (uniform circular motion), the *successor* is $T_{\mathrm{new}} := \{x_0 = a\} \cup \{x_{t+1} = A x_t : t < \bar{t}\}$, the *honest old law* is $T^{\varepsilon} := \{|x_t - \mathrm{Rot}^t a| \le \varepsilon : t \le \bar{t}\}$, and a *time-dependent form* is $T^{(\varepsilon_\cdot)} := \{|x_t - \mathrm{Rot}^t a| \le \varepsilon_t : t \le \bar{t}\}$.

Proposition A.37 (Explanation by approximate correction)**.**

(a) $T_{\mathrm{new}} \vdash T^{\varepsilon}$ *for every rational* $\varepsilon \ge \varepsilon(\bar{t}) := ((1 + \eta_c \|\mathbf{P}\|)^{\bar{t}} - 1)|a|$.

(b) $T_{\mathrm{new}} \vdash T^{\varepsilon}$ *iff* $\varepsilon^2 \ge \varepsilon_{\min}(\bar{t})^2 := \max_{t \le \bar{t}} |A^t a - \mathrm{Rot}^t a|^2$*, a rational number. So* $\varepsilon_{\min}(\bar{t}) \le \varepsilon(\bar{t})$*, and* $T^{\varepsilon_{\min}(\bar{t})}$ *is the strongest uniform-tolerance form of the old law on the window that the successor proves.*

(c) If $\eta_c \mathbf{P} a \ne 0$ *and* $\bar{t} \ge 1$*, then* $T_{\mathrm{new}} \vdash \neg \bigwedge T_0$.

(d) Among uniform-tolerance forms, T^{ε} *is a common consequence of* T_0 *and* T_{new} *iff* $\varepsilon \ge \varepsilon_{\min}(\bar{t})$*; the strongest common one is* $T^{\varepsilon_{\min}(\bar{t})}$.

(e) The time-dependent form with $\varepsilon_t := |A^t a - \mathrm{Rot}^t a|$ *is a common consequence of both theories, strictly stronger than* $T^{\varepsilon_{\min}(\bar{t})}$ *whenever* $\varepsilon_t < \varepsilon_{\min}(\bar{t})$ *for some* t.

(f) The strongest common consequence of the two finite theories is their disjunction $\bigwedge T_0 \vee \bigwedge T_{\mathrm{new}}$*, which is not a tolerance law; if* $\varepsilon_{\min}(\bar{t}) > 0$*,* $T^{\varepsilon_{\min}(\bar{t})}$ *does not entail it.*

Proof. Write $\varepsilon_t := |A^t a - \mathrm{Rot}^t a|$; ε_t^2 is rational because $A^t a$ and $\mathrm{Rot}^t a$ are rational vectors. In every model of $B \cup T_{\mathrm{new}}$, that is in any real closed field with interpretations of the constants, the recurrence determines $x_t = A^t a$, a rational vector, since $\mathbb{Q}$ embeds uniquely into every ordered field; instantiating the recurrence t times gives $T_{\mathrm{new}} \vdash x_t = A^t a$.

(a) $\|\mathrm{Rot}\| = 1$ and $\|A\| \le 1 + \eta_c \|\mathbf{P}\|$, so $\varepsilon_{t+1} = |A(A^t a - \mathrm{Rot}^t a) + \eta_c \mathbf{P}\, \mathrm{Rot}^t a| \le (1 + \eta_c \|\mathbf{P}\|)\varepsilon_t + \eta_c \|\mathbf{P}\| |a|$ with $\varepsilon_0 = 0$, and by induction $\varepsilon_t \le ((1 + \eta_c \|\mathbf{P}\|)^t - 1)|a| \le \varepsilon(\bar{t})$ for $t \le \bar{t}$. Since T_{new} proves $|x_t - \mathrm{Rot}^t a|^2 = \varepsilon_t^2$, it proves each conjunct of T^{ε} for rational $\varepsilon \ge \varepsilon(\bar{t})$.

(b) Over $B \cup T_{\mathrm{new}}$ each conjunct of T^{ε} is equivalent to the inequality $\varepsilon_t^2 \le \varepsilon^2$ between rationals, which RCF proves or refutes. So $T_{\mathrm{new}} \vdash T^{\varepsilon}$ iff all $\bar{t} + 1$ inequalities hold, iff $\varepsilon^2 \ge \max_{t \le \bar{t}} \varepsilon_t^2 = \varepsilon_{\min}(\bar{t})^2$; and $\varepsilon_{\min}(\bar{t}) \le \varepsilon(\bar{t})$ by the bound in (a).

(c) $T_{\mathrm{new}} \vdash x_1 = Aa$ and $T_0 \ni (x_1 = \mathrm{Rot}\, a)$, with $Aa - \mathrm{Rot}\, a = \eta_c \mathbf{P} a \ne 0$.

(d) T_0 proves $x_t = \mathrm{Rot}^t a$, hence T^{ε} for every $\varepsilon \ge 0$; combine with (b).

(e) T_{new} proves $|x_t - \mathrm{Rot}^t a|^2 = \varepsilon_t^2$, and T_0 proves $|x_t - \mathrm{Rot}^t a|^2 = 0 \le \varepsilon_t^2$. If $\varepsilon_t < \varepsilon_{\min}(\bar{t})$ at some t, choose a real point x_t with $\varepsilon_t < |x_t - \mathrm{Rot}^t a| \le \varepsilon_{\min}(\bar{t})$ and put $x_s := \mathrm{Rot}^s a$ for $s \ne t$: this is a model of $T^{\varepsilon_{\min}(\bar{t})}$ that violates the time-dependent form.

(f) For finite theories, a sentence follows from both iff it follows from their disjunction (proof by cases), so the disjunction is the strongest common consequence. Put $x_t := \mathrm{Rot}^t a + (\varepsilon_{\min}(\bar{t}), 0)$ for every $t \le \bar{t}$: if $\varepsilon_{\min}(\bar{t}) > 0$, this is a model of $T^{\varepsilon_{\min}(\bar{t})}$ with $x_0 \ne a$, so it is a model of neither theory, and $T^{\varepsilon_{\min}(\bar{t})}$ does not entail the disjunction; in particular the disjunction is not equivalent to a uniform-tolerance law. □

Remark A.38 (Derivations of polynomial size)*. Proof sketch.* What is not written out is the size of derivations of arithmetic on binary numerals from the field axioms, which is standard. The

natural derivation of T^{ε} from T_{new} instantiates the recurrence t times, writes each x_t as an exact rational vector and checks $\bar{t}+1$ numerical inequalities; the numerals it writes have total bit size $\sum_{t\le\bar{t}} \text{bits}(A^t a) = O(\bar{t}^2 \log \text{ht})$, where ht is the height of the entries of A. If the numeral arithmetic has derivations of size polynomial in the numerals' size, the whole derivation has size polynomial in $\bar{t}$. For $\mathbf{P} = \mathbf{J}\,\text{Rot}$, with $\mathbf{J}$ the rotation by 90°, the numerals have 1894 bits at $\bar{t} = 10$ and about 1.8×10^5 at $\bar{t} = 100$.

Exact rational arithmetic (code in the accompanying repository) gives, for three perturbations with $\|\mathbf{P}\| = 1$ ($\mathbf{J}$ is the rotation by 90°):

$\mathbf{P}$	$\varepsilon_{\min}(\bar{t})$ at $\bar{t} = 1, 10, 100, 200$	bound $\varepsilon(\bar{t})$
$\mathbf{J}\,\text{Rot}$ (outward spiral with precession)	0.0100, 0.0999, 0.961, 1.69	0.0100, 0.105, 1.70, 6.32
$\text{diag}(1,-1)\,\text{Rot}$ (distortion)	0.0100, 0.0127, 0.0186, 0.0243	the same
identity (spiral)	0.0100, 0.103, 1.33, 3.49	the same

In each case T_{new} refutes T_0 at $t = 1$. The bound $\varepsilon(\bar{t})$ is nearly tight for small windows when the perturbation accumulates, and very loose for a bounded distortion (ratio 0.004 at $\bar{t} = 200$). For $\mathbf{P} = \mathbf{J}\,\text{Rot}$ the time-dependent tolerances have mean 0.0500 at $\bar{t} = 10$ and 0.490 at $\bar{t} = 100$, about half of $\varepsilon_{\min}(\bar{t})$. A circle refitted on the window (radius and angular speed; floating point, for comparison only) absorbs the precession almost entirely in the first case (2.5×10^{-4} and 2.5×10^{-3} at $\bar{t} = 10, 100$) and the growth not at all in the third (3.7×10^{-2} and 0.80): the toy analogue of the fitted ellipses of Table 1.

A.8 Domains of applicability and retention

Let a law Λ with per-system elements prove the data of every system but one, $\star$, whose $n_\star$ data it fits, with $\star$'s elements ($k_\star$ bits), at the per-datum rate $m^\star_{\text{pat}}$ of the exact law with exceptions, or at the rate $m^\star_{\text{hon}}$ of an honest tolerance for $\star$, at a one-off cost of $h_\star$ bits. The *domain-restricted law* $\Lambda_{\text{Dom}} := \forall b\,(\text{Dom}(b) \to \text{law}(b))$, with Dom a definable predicate that B proves of the other systems and refutes of $\star$, costs $\|\text{Dom}\|$ bits more than Λ (a few bits if "planet" is a base predicate, about the logarithm of the number of systems to name $\star$) and leaves $\star$'s data literal, at rate γ. So $\star$'s data have three treatments: restriction (rate γ, constant $\lambda\|\text{Dom}\|$), patched (rate $m^\star_{\text{pat}}$, constant $\lambda k_\star$) and honest (rate $m^\star_{\text{hon}}$, constant $\lambda(k_\star + h_\star)$).

Proposition A.39 (Domain restriction)**.**

(a) $J_\lambda(\Lambda_{\text{Dom}}) - J_\lambda(\Lambda \cup \text{El}_\star$ *with* $\star$ *'s exceptions*$) = \lambda\|\text{Dom}\| - \lambda k_\star + \sum_{\star\text{'s data}}(\gamma - c_{\text{pat}})$*, and likewise against the honest treatment of* $\star$*, with* $\lambda(k_\star + h_\star)$ *and* c_{hon}*, where* $c_{\text{pat}}, c_{\text{hon}}$ *are the per-datum costs.*

(b) With i.i.d. data and $n_\star \to \infty$*, the restriction is eventually the* λ*-MAP's treatment of* $\star$ *iff* $\gamma < \min(m^\star_{\text{pat}}, m^\star_{\text{hon}})$*, or* $\gamma = m^\star_{\text{hon}} < m^\star_{\text{pat}}$ *with the honest cost almost surely* γ *and* $\|\text{Dom}\| < k_\star + h_\star$*. If the least rate is shared by two treatments whose costs differ non-degenerately, there is no eventual winner.*

(c) Under SAI's pricing (code R), $m^\star_{\text{hon}} < \gamma$ *whenever* $\varepsilon^*_\star < \mathsf{R}/(2\mathrm{e})$ *(for* $\Delta \le 2\varepsilon^*_\star$*) or* $\Delta < \mathsf{R}/\mathrm{e}$ *(for* $\Delta \ge 2\varepsilon^*_\star$*). So a system that the law confines within about* 66° *of longitude (a sufficient bound) is never cut out asymptotically at any cell width below* R/e*: it is kept with its own, wider tolerance. Under code Q with* $\Delta \le 2\varepsilon^*_\star$ *the honest treatment costs exactly* γ *per datum, and the restriction is eventually selected iff* $m^\star_{\text{pat}} > \gamma$ *and* $\|\text{Dom}\| < k_\star + h_\star$*.*

Proof. (a) The two theories differ in the domain clause, in $\star$'s elements, and in the treatment of $\star$'s data; the other systems' data are proved by both. (b) Theorem A.22(i), (ii) applied to the three treatments (code-Q costs are bounded and code-R costs have all moments): a unique least rate wins eventually; a tie with an almost surely zero cost difference is decided by the constants, which is the case of the restriction and the honest treatment under code Q in the fine regime, where the honest law is silent on every datum and costs exactly γ; and a tie with a non-degenerate difference has no eventual winner. (c) In the fine regime Lemma A.20(ii) gives $m^{\star}_{\mathrm{hon}} \le \log(2\varepsilon^{*}_{\star}/\Delta) + \log \mathrm{e}$, which is below $\log(\mathsf{R}/\Delta)$ iff $2\mathrm{e}\varepsilon^{*}_{\star} < \mathsf{R}$. In the coarse regime $m^{\star}_{\mathrm{hon}} = s_{\Delta}\beta_e \le \log \mathrm{e}$, which is below $\log(\mathsf{R}/\Delta)$ iff $\Delta < \mathsf{R}/\mathrm{e}$. Under code Q with $\Delta \le 2\varepsilon^{*}_{\star}$ the honest law is silent on every datum (Lemma A.15), and (b) applies. □

For the Moon over one year, with one fixed ellipse ($\varepsilon^{*} = 156.3' = 2.6^{\circ}$, $\mathbb{E}|e| = 61.7'$ for the mean fit, $k_{\star} = 60$, $h_{\star} = 20$, $\|\mathrm{Dom}\| = 4$, $c_{\mathrm{time}} = 10.5$), an exact computation (code in the accompanying repository) gives the eventually selected treatment of the Moon's data, with rates in bits per datum ("excluded" is the domain restriction, chosen at an exact tie between the honest and the literal treatment):

Δ	γ	code Q, $\lambda = 1$	code Q, $\lambda = 2$	code R, $\lambda = 1, 2$
$10'$	11.08	excluded (tie; patched 20.52)	excluded (patched 30.51)	honest (4.973)
$60'$	8.49	excluded (tie; patched 13.66)	excluded (patched 21.22)	honest (2.445)
$300'$	6.17	patched (3.430)	patched (5.592)	honest (0.686)
$600'$	5.17	patched (1.612; honest 2.694)	patched (2.6933; honest 2.6939)	honest (0.343)
$1800'$	3.58	patched (0.483; honest 0.623)	honest (0.623; patched 0.843)	honest (0.114)

At the exact ties the constants decide ($\|\mathrm{Dom}\| = 4$ against $k_{\star} + h_{\star} = 80$). In code R the honest treatment overtakes the restriction after about $\lambda \cdot 76/(\gamma - m^{\star}_{\mathrm{hon}})$ of the Moon's data in expectation, about 12 at $\lambda = 1$ and 25 at $\lambda = 2$ at $\Delta = 10'$ and $60'$. More generally, for every planet and window of Table 1, and for the Moon over a month, a year and 18.6 years, a law with its own tolerance has a lower rate under code R than leaving the body's data literal at every cell width computed, by Proposition A.39(c); a sharp boundary is selected only when even an honest tolerance is worse than literal data, or under literal pricing at fine precision.

For retention, let a datum proved from T cost in addition $r_{\mathrm{d}}\,\mathrm{dl}_T$ bits, where dl_T is the size of the derivation used, and let $J^{\mathrm{der}}_{\lambda}(T; D) := \lambda\|T\| + \mathrm{Exc}_D(T) + \sum_{\text{data proved}} r_{\mathrm{d}}\,\mathrm{dl}_T$. Compare an honest old law T_{old} (tolerance $\varepsilon_{\mathrm{old}}$ on a window) with an exact successor T_{new} that proves every datum, with per-datum derivation charges $c_{\mathrm{old}} < c_{\mathrm{new}}$.

Proposition A.40 (Retention of a superseded theory)**.**

(a) T_{old} *beats* T_{new} *on* D *iff*

$$\lambda(\|T_{\mathrm{new}}\| - \|T_{\mathrm{old}}\|) + (c_{\mathrm{new}} - c_{\mathrm{old}})\,N_{\mathrm{p}} + c_{\mathrm{new}}\,(n - N_{\mathrm{p}}) > \mathrm{Exc}_D(T_{\mathrm{old}}),$$

where N_{p} *is the number of data that* T_{old} *proves and* $n = |D|$*.*

(b) With i.i.d. data the rates are $m_{\mathrm{old}} = s_{\mathrm{old}}\gamma_{\mathrm{si}} + (1 - s_{\mathrm{old}})c_{\mathrm{old}}$*, with* s_{old} *the silence probability of* T_{old}*, and* $m_{\mathrm{new}} = c_{\mathrm{new}}$*. If* $m_{\mathrm{old}} < m_{\mathrm{new}}$ *(respectively* $>$*), then almost surely* T_{old} *(respectively* T_{new}*) is eventually the* λ*-MAP of the two, whatever* λ *and the lengths are. At* $m_{\mathrm{old}} = m_{\mathrm{new}}$ *the lengths decide if the cost difference is almost surely constant, and otherwise there is no eventual winner.*

(c) *Let* $s_{\mathrm{old}} = \min(2\varepsilon_{\mathrm{old}}/\Delta, 1)$. *Under code Q,* m_{old} *is non-increasing in* Δ *on* $\{\Delta : \gamma(\Delta) \geq c_{\mathrm{old}}\}$*; under code R it is non-increasing in* Δ *on* $(0, \infty)$ *if* $\beta_e \geq c_{\mathrm{old}}$. *In either case the cell widths in that range at which* T_{old} *is retained form an interval with no upper bound inside the range; call its lower end* Δ^*. *For* $\Delta \geq 2\varepsilon_{\mathrm{old}}$ *under code Q,* Δ^* *solves* $2\varepsilon_{\mathrm{old}}(\gamma(\Delta) - c_{\mathrm{old}})/\Delta = c_{\mathrm{new}} - c_{\mathrm{old}}$.

Proof. (a) $J_\lambda^{\mathrm{der}}(T_{\mathrm{new}}) = \lambda\|T_{\mathrm{new}}\| + c_{\mathrm{new}} n$ and $J_\lambda^{\mathrm{der}}(T_{\mathrm{old}}) = \lambda\|T_{\mathrm{old}}\| + \mathrm{Exc}_D(T_{\mathrm{old}}) + c_{\mathrm{old}} N_{\mathrm{p}}$; compare term by term. (b) T_{old} is honest, so it never refutes a datum; a datum costs c_{old} when it proves it and γ_{si} when it is silent. Apply Theorem A.22. (c) *Code Q.* In the coarse regime $m_{\mathrm{old}} = c_{\mathrm{old}} + s_{\mathrm{old}}(\gamma(\Delta) - c_{\mathrm{old}})$, and s_{old} and $\gamma(\Delta) - c_{\mathrm{old}} \geq 0$ are both non-negative and non-increasing in Δ, so their product is too; in the fine regime $m_{\mathrm{old}} = \gamma(\Delta)$, which is decreasing. *Code R.* In the fine regime T_{old} is silent on every datum and $m_{\mathrm{old}} = \mathbb{E}\bar{c}_\Delta(w)$, strictly decreasing by Lemma A.20(iii); in the coarse regime $m_{\mathrm{old}} = c_{\mathrm{old}} + s_{\mathrm{old}}(\beta_e - c_{\mathrm{old}})$, non-increasing when $\beta_e \geq c_{\mathrm{old}}$. The two forms agree at $\Delta = 2\varepsilon_{\mathrm{old}}$. A non-increasing rate stays below c_{new} once it is below it, which gives the interval, and the equation for Δ^* is $m_{\mathrm{old}} = c_{\mathrm{new}}$ in the coarse regime under code Q. □

With derivations from the successor assumed a thousand times larger ($\mathrm{dl}_{T_{\mathrm{new}}}/\mathrm{dl}_{T_{\mathrm{old}}} = 1000$, as for a validated integration against solving Kepler's equation), exact code-R costs and bisection (code in the accompanying repository) give the least retained cell width Δ^* for $c_{\mathrm{new}} = 0.01$, 0.1 and 1 bit per datum:

body, window	ε^*	code Q	code R
Mercury, 100 yr	2.02′	1542′, 258.5′, 37.2′	229.8′, 22.97′, 2.33′
Mars, 12 yr	1.44′	1202′, 195.6′, 27.7′	174.2′, 17.42′, 1.744′
Mars, 2000 yr	49.9′	10450′, 2895′, 533′	5717′, 572′, 57.8′
Jupiter, 100 yr	6.57′	3468′, 661′, 101.6′	867′, 86.7′, 8.61′
Saturn, 2000 yr	81.9′	never, 3994′, 784′	10320′, 1032′, 102.9′
Moon, 1 month	30.9′	8414′, 2086′, 364′	4473′, 447′, 44.4′

Under random completion the retention region reaches precisions 1.8 to 6.9 times finer than under literal pricing at $c_{\mathrm{new}} = 0.01$, 3.9 to 11 times at 0.1, and 7.6 to 16 times at 1, because the honest law's silence is cheap.

Question A.41. What is the posterior over an explicit pool of domain predicates (base predicates such as "planet", windows, and their conjunctions, with stated syntax costs) on the Kepler residuals? By Proposition A.39 we expect the answer to depend on the pricing: tolerances with error models under SAI's pricing, and sharp domains under literal pricing at fine precision.

A.9 What log loss does and does not measure

This subsection states and proves the results behind Section 1.3. Losses are in bits, and for every score lower is better. For a real random variable X, $\mathrm{h}(X) := -\int f \log f$ is the differential entropy of its density f (possibly $-\infty$), and $\mathrm{h}(X) := -\infty$ if X has no density; H is the Shannon entropy and H_2 the binary entropy. For a forecast P of a real quantity and an outcome x, $\mathrm{CRPS}(P, x) := \int (P(X \leq t) - \mathbf{1}[x \leq t])^2 \, dt$ is the continuous ranked probability score, a proper score that grows with the distance of P's mass from x (cf. Gneiting and Raftery, 2007). In regime R we call the rate $r_{\mathcal{F}}(\Delta)$ of Proposition 1.2(c), the expected cost per datum of $\mathrm{BC}_{\mathcal{F}}$ when the errors have the law that $\mathcal{F}$ determines, the *price* of the calibrated completion of a law with that error law.

Proposition A.42 (What questions about one quantity encode)**.** *Let $B \supseteq$ RCF have a constant q; for a finite union C of rational intervals let σ_C be the sentence "$q \in C$". For a theory machine G let $\beta_G(C) := \nu_G(\{(\sigma_C, 1)\})$, the probability that G's random theory is consistent and proves $q \in C$, and define the cumulative loss of G along a reveal sequence as in the semimeasure convention of* **??***, with ν_G in place of M_A.*

(a) Let D be a two-valued data set of sentences σ_C with $\nu_G(D) > 0$, and C_D the Boolean combination of their sets that the labels determine. Then $B \vdash \psi_D \leftrightarrow \sigma_{C_D}$, $\nu_G(D) = \beta_G(C_D)$, and on every consistent reveal sequence of D, in any order and with repeats, the cumulative loss of G is $-\log \beta_G(C_D) + \log \nu_G(\emptyset)$. If D answers every question of a finite family of interval questions about q truly, C_D is the cell of the true value in the partition that their intervals generate, and β_G is a sub-probability on the cells of that partition: two machines that give the true cell the same mass pay the same, wherever the rest of their mass lies.

(b) Let the true value be 0, let $C = [a_0, a_0 + \Delta)$ be a fixed cell containing it, and let P be a probability law of q. For $P = N(d, \sigma^2)$, $-\log P(C) = d^2/(2\sigma^2 \ln 2) + O(|d|)$ as $|d| \to \infty$; for the Laplace law with location d and scale b, $-\log P(C) = |d|/(b \ln 2) + O(1)$. The same holds, averaged over the dither, for the cell of 0 in a uniformly dithered grid of width Δ.

(c) Let q live on a circle of circumference R (an integer multiple of Δ, at least 2Δ), let the question be the cell of q in a grid of width Δ with a uniform offset, and let $P_{d,s} := (1 - s)\delta_d + s\,U$ with U uniform on the circle and $0 < s < 1$. With x the true value and $\operatorname{sep}(x, d) := \min(|x - d|/\Delta, 1)$, $|x - d|$ the distance on the circle, the expected loss of $P_{d,s}$ over the offset is $\operatorname{sep}(x, d) \log(\mathsf{R}/(s\Delta)) + (1 - \operatorname{sep}(x, d))(-\log(1 - s + s\Delta/\mathsf{R}))$.

Proof. (a) RCF proves the Boolean identity, and $\nu_G(D)$ depends on D only through $\mathrm{Cn}(D^{\pm})$ (**??**(c)), so $\nu_G(D) = \beta_G(C_D)$. ν_G is monotone ($D' \subseteq D$ gives $E^G_{D'} \supseteq E^G_D$), so every prefix D_{t-1} has $\nu_G(D_{t-1}) \geq \nu_G(D) > 0$; the conditional of a new datum is $\nu_G(D_t)/\nu_G(D_{t-1})$ and that of a repeat is 1 (**??**), and the product telescopes to $\nu_G(D)/\nu_G(\emptyset)$. True answers to all the questions determine the cell of the true value. Distinct cells are B-incompatible and a consistent theory proves at most one of them, so the events defining β_G on the cells are disjoint. (b) Let $d \to +\infty$ and put $z_1 := (d - a_0 - \Delta)/\sigma < z_2 := (d - a_0)/\sigma$, so that $P(C) = \Phi(-z_1) - \Phi(-z_2)$ with Φ the standard normal distribution function. Since $z_2 - z_1 = \Delta/\sigma$ is fixed, $\Phi(-z_2)/\Phi(-z_1) \to 0$, and the Mills-ratio asymptotics $\ln \Phi(-z) = -z^2/2 - \ln z + O(1)$ give $-\ln P(C) = (d - a_0 - \Delta)^2/(2\sigma^2) + O(\ln d) = d^2/(2\sigma^2) + O(d)$, uniformly for a_0 in a bounded set. For the Laplace law and $d \geq a_0 + \Delta$, $P(C) = \frac{1}{2} e^{-(d - a_0 - \Delta)/b}(1 - e^{-\Delta/b})$. The case $d \to -\infty$ is symmetric, and the bounds are uniform in the offset, so they survive averaging. (c) x and d lie in different cells iff a grid point falls on the shorter arc between them, which has probability $\operatorname{sep}(x, d)$. If they do, $P_{d,s}$ gives the cell of x the mass $s\Delta/\mathsf{R}$; otherwise $1 - s + s\Delta/\mathsf{R}$. □

So under log loss misses are graded, without bound, but by each hypothesis's own error model; a family of questions about one quantity fixes only a resolution, and asking each of many values one randomly offset question grades near-point misses by sep, a bounded metric that is constant beyond one cell width. Regime R is such a design. A direct computation (code in the accompanying repository) gives, for Gaussian forecasts with $\sigma = 1$ and $\Delta = 0.1$, an extra cost of 25.95 bits at $d = 6$ over $d = 0$, against $d^2/(2 \ln 2) = 25.97$, on a fixed and on a dithered grid alike. For a single yes/no datum every scoring rule is local; for three or more outcomes see **??**.

Proposition A.43 (Fitted error models rank laws by error norms)**.** *Let $e_1, \ldots, e_n$ be the errors of a law on n data, $M_p := (\frac{1}{n} \sum_j |e_j|^p)^{1/p}$ and $M_\infty := \max_j |e_j|$. With the scale fitted by maximum*

likelihood, the least total log loss of the error densities is $\frac{n}{2}\log(2\pi\mathrm{e}M_2^2)$ *for a Gaussian error model,* $n\log(2\mathrm{e}M_1)$ *for a Laplace one,* $n\log(2M_\infty)$ *for the uniform law on* $[-\delta,\delta]$ *(at* $\delta = M_\infty$*; for smaller* δ *the loss is infinite), and* $n\log\big(2\Gamma(1+\frac{1}{p})(p\mathrm{e})^{1/p}M_p\big)$ *for the density proportional to* $\exp(-|z/s|^p)$*. So these error models rank laws by RMS, mean absolute, maximum and* L^p *error.*

Proof. In nats the total loss of the last family is $nM_p^p/s^p + n\ln\big(2s\Gamma(1+\frac{1}{p})\big)$, minimised at $s^p = pM_p^p$; $p=2$ and $p=1$ are the Gaussian and Laplace cases. The uniform case is monotone in $\delta \geq M_\infty$. □

Lemma A.44 (Dithered cells)**.** *Let* e *be a real random variable,* $\tilde{u}$ *uniform on* $(0,1]$ *and* u *uniform on* $[0,1)$*, both independent of* e*, and* $C_u(e)$ *the cell of the grid* $\{(k-u)\Delta : k\in\mathbb{Z}\}$ *that contains* e*. Then* $H(C_u(e)\mid u) = \mathrm{h}(e+\Delta\tilde{u}) - \log\Delta$*. Here* $e+\Delta\tilde{u}$ *has the density* $f(y) = \mathbb{P}[y-\Delta\leq e<y]/\Delta \leq 1/\Delta$*, so* $\mathrm{h}(e+\Delta\tilde{u}) \geq \log\Delta$*.*

Proof. Given u, the cell $[(k-u)\Delta,(k+1-u)\Delta)$ has probability $\Delta f((k+1-u)\Delta)$. Average $\sum_k -\Delta f\log(\Delta f)$ at these points over u; the substitution $y=(k+1-u)\Delta$ turns the average into $\int -f(y)\log(\Delta f(y))\,dy = \mathrm{h}(e+\Delta\tilde{u}) - \log\Delta$ (Tonelli's theorem, all terms being non-negative). □

This is the known rate of entropy-coded subtractively dithered quantisation; the short proof is included for completeness.

Theorem A.45 (The best rate in regime R is a smoothed entropy)**.** *In regime R, with* $\mathsf{R} \geq 2\varepsilon^* + \Delta$*, let* $\varepsilon \geq \varepsilon^*$ *and* $\mathcal{F}$ *be as in Proposition 1.2, and* $\tilde{u}$ *as in Lemma A.44. Then* $r_{\mathcal{F}}(\Delta) = \mathrm{h}(e+\Delta\tilde{u}) - \log\Delta$*.*

(a) If the support of e *lies in an interval of length at most* Δ*, then* $r_{\mathcal{F}}(\Delta) = \mathrm{disp}(e)/\Delta$*, where* $\mathrm{disp}(e) := \int H_2(F_e(t))\,dt = 2\varepsilon c_{\mathcal{F}}$ *and* F_e *is the distribution function of* e*;* disp *is translation invariant and* $\mathrm{disp}(\lambda e) = |\lambda|\,\mathrm{disp}(e)$*.*

(b) As $\Delta\to 0$*,* $r_{\mathcal{F}}(\Delta) - \log(1/\Delta) \to \mathrm{h}(e) \in [-\infty,\infty)$*.* $\mathrm{h}(e)$ *is invariant under measure-preserving rearrangements of the density of* e*, and* $\mathrm{h}(\lambda e) = \mathrm{h}(e) + \log|\lambda|$*.*

(c) If e *takes finitely many values,* $r_{\mathcal{F}}(\Delta) = H(e)$ *for every* Δ *up to the least gap between them. If the law of* e *has a non-atomic part,* $r_{\mathcal{F}}(\Delta)\to\infty$ *as* $\Delta\to 0$*.*

Proof. By the proof of Proposition 1.2(c), $r_{\mathcal{F}}(\Delta)$ is the conditional entropy of the answer given the question. Given the time t_j and the dither, the answer is the cell of $q(t_j) = f(t_j) - e_j$; translation by $f(t_j)$ keeps the dither uniform, and $\mathsf{R}\geq 2\varepsilon^*+\Delta$ keeps the cells that meet the band distinct on the circle, so $r_{\mathcal{F}}(\Delta) = H(C_u(-e)\mid u) = \mathrm{h}(-e+\Delta\tilde{u}) - \log\Delta$ by Lemma A.44. Since $-e+\Delta\tilde{u} = -(e+\Delta(1-\tilde{u}))+\Delta$ and $1-\tilde{u}$ has the law of $\tilde{u}$ up to a null set, this is $\mathrm{h}(e+\Delta\tilde{u})-\log\Delta$, and also $H(C_u(e)\mid u)$. (a) With the support in $[a,b]$, $b-a\leq\Delta$, the density of $e+\Delta\tilde{u}$ is $F_e(y^-)/\Delta$ on $[a,b]$, $1/\Delta$ on $[b,a+\Delta]$ and $(1-F_e((y-\Delta)^-))/\Delta$ on $[a+\Delta,b+\Delta]$; integrating $-f\log(\Delta f)$ gives $\frac{1}{\Delta}\int_a^b H_2(F_e(t))\,dt$. The substitution $t=\varepsilon-2\varepsilon v$ and $H_2(1-x)=H_2(x)$ give $\mathrm{disp}(e) = 2\varepsilon c_{\mathcal{F}}$. (b) *Lower bound.* If e has a density, $\mathrm{h}(e+\Delta\tilde{u}) \geq \mathrm{h}(e+\Delta\tilde{u}\mid\tilde{u}) = \mathrm{h}(e)$; otherwise $\mathrm{h}(e) = -\infty$. *Upper bound.* Fix a compact interval I of length ℓ that contains the support of $e+\Delta\tilde{u}$ for every $\Delta\leq 1$, and let Q be the uniform law on I. For a law P on I, $\mathrm{h}(P) = \log\ell - \mathrm{KL}(P\|Q)$, with KL $=\infty$ when P has no density. As $\Delta\to 0$, $e+\Delta\tilde{u}\to e$ weakly, and relative entropy is lower semicontinuous under weak convergence (by the Donsker–Varadhan formula it is a supremum, over bounded continuous g, of the continuous functions $P\mapsto \mathbb{E}_P g - \log\mathbb{E}_Q 2^g$). So $\limsup_{\Delta\to 0}\mathrm{h}(e+\Delta\tilde{u}) \leq \log\ell - \mathrm{KL}(e\|Q) = \mathrm{h}(e)$. The invariances hold because h depends on the density only through the measures of its level sets. (c) With

atoms p_i at least Δ apart, the density of $e + \Delta\tilde{u}$ is p_i/Δ on disjoint intervals of length Δ, so $\mathrm{h}(e+\Delta\tilde{u}) = H(e) + \log\Delta$. If the law of e has a non-atomic part of mass $w_\mathrm{c} > 0$, let Z indicate it. Then $H(C_u(e) \mid u) \geq H(C_u(e) \mid u, Z) \geq w_\mathrm{c}\,\mathbb{E}_u[-\log\max_k p_k^\mathrm{c}(u)] \geq w_\mathrm{c}\log(1/Q_\mathrm{c}(\Delta))$, where $p_k^\mathrm{c}(u)$ are the cell probabilities of the normalised non-atomic part and $Q_\mathrm{c}(\Delta) := \sup_a P_\mathrm{c}([a, a+\Delta))$, which tends to 0 as $\Delta \to 0$ for a non-atomic law. □

So in the coarse regime the pricing is a dispersion of the error: sensitive to distance within one cell width, and blind to a constant bias, which the calibrated error model absorbs. In the fine limit it is the entropy of the error: sensitive to the volume the errors occupy and blind to where that volume lies; and at fine enough resolution a law whose error takes finitely many values beats every law whose error law has a non-atomic part, however small its errors. The convergence in (b) can be slow: for the sinusoidal error of Proposition A.13 with amplitude 1 (the arcsine law on $[-1, 1]$, whose density is unbounded), a direct computation (code in the accompanying repository) gives $r_\mathcal{F}(\Delta) - \log(1/\Delta) - \mathrm{h}(e) = 0.8542$, 0.1935, 0.0274 and 8.66×10^{-4} bits at $\Delta = 1$, 0.05, 10^{-3} and 10^{-6}, decaying about as $0.866\sqrt{\Delta}$. Part (b) of Proposition 1.2 does not depend on the logarithmic score: the errors are independent of the questions, so the cell law of $\mathrm{BC}_\mathcal{F}$ is the true conditional law of the answer given the question, which every proper score rewards at least as much as any other band-confined completion. What is specific to log loss is the ranking across laws.

Proposition A.46 (Rival laws in regime R)**.** *Let f_A and f_B be laws for one quantity q on a circle of circumference R, whose errors $e_A(t) = f_A(t) - q(t)$ and $e_B(t) = f_B(t) - q(t)$ lie in arcs shorter than $\mathsf{R}/2$. If the errors of f_A at the data times are i.i.d., with law $\mathcal{F}_A$ and independent of the questions, and the errors of f_B at the data times are identically distributed, then $f_A(t_j) - f_B(t_j)$ is the same for every j (modulo R), and the two laws' calibrated completions have the same price at every cell width.*

Proof. $e_B(t_j) = e_A(t_j) - d_j$ with $d_j := f_A(t_j) - f_B(t_j)$, so $\mathcal{F}_A$ rotated by d_j is the law of $e_B(t_j)$ for every j, and $\mathcal{F}_A$ is invariant under rotation by $d_j - d_1$. A law on an arc shorter than half the circle is invariant under no non-zero rotation: invariance under a rotation by an irrational fraction of R would make its closed support the whole circle, and invariance under a rotation by a rational fraction p/k in lowest terms, $k \geq 2$, gives invariance under rotation by R/k, so that the support contains k points spaced R/k apart, which lie in no arc shorter than $\mathsf{R} - \mathsf{R}/k \geq \mathsf{R}/2$. So $d_j = d_1$, and the prices agree because h and disp are translation invariant. □

So when two rival laws are compared by the prices of their calibrated completions, at least one of them has errors that are not random in the sense of regime R, and SAI's own rate on such data can be lower than either price; if one law's errors are random, its calibrated completion is never out-priced (Proposition 1.2(c)).

Example A.47 (Rival laws, priced)**.** (a) Law A has errors uniform on $[-1, 1]$. Law B has errors $-M$, 0, M with probabilities 0.1, 0.8, 0.1, $M = 1000$: exactly right on 80% of the times and off by 1000 otherwise. By Theorem A.45 the prices of their calibrated completions are $r_A(\Delta) = 1 + \Delta/(4\ln 2) - \log\Delta$ for $\Delta \leq 2$ and $(\log \mathrm{e})/\Delta$ for $\Delta \geq 2$, and $r_B(\Delta) = H(0.1, 0.8, 0.1)$ for $\Delta \leq M$ and $2MH_2(0.1)/\Delta$ for $\Delta \geq 2M$. So B is priced below A exactly for the cell widths below the root Δ_c of $r_A(\Delta) = H(0.1, 0.8, 0.1)$. A direct computation (code in the accompanying repository) gives $H(0.1, 0.8, 0.1) = 0.92193$ bits and $\Delta_c = 1.558542$. Yet the maximum, RMS and mean absolute errors (1, 0.577 and 0.5 for A; 1000, 447.2 and 200 for B), the fitted error models of

Proposition A.43 and the CRPS of the calibrated forecasts (0.333 against 180) prefer A at every resolution.

(b) Law B' has error $s(t) \in \{-1, 1\}$, a fair sign, and law A' has error $s(t)U(t)$, with $U(t)$ uniform on $[0, 1]$ and independent of $s(t)$. At every time the prediction of A' lies between the truth and that of B'. The price of B''s calibrated completion is 1 bit for $\Delta \leq 2$ and that of A' is $r_A(\Delta)$, so B' is priced below A' exactly for Δ below the smaller root Δ'_c of $\Delta = 4 \ln \Delta$, and a direct computation gives $\Delta'_c = 1.429612$. The calibrated CRPS prefers A' (1/3 against 1/2).

(c) On a circle of circumference $\mathsf{R} \geq 4M$ let Z be uniform and independent of (e_A, e_B), with e_A and e_B independent and distributed as in (a); put $f_A := Z$, $q := f_A - e_A$ and $f_B := q + e_B$. Then $f_B = Z - e_A + e_B$ is uniform and independent of (e_A, e_B), so e_A is independent of f_A and e_B of f_B, and each calibrated forecast is the conditional law of q given that law's own prediction: an ideal forecast for its own information. The two information sets are not nested, and the pair determines q, since e_B is the element of $\{-M, 0, M\}$ nearest to $f_B - f_A$ (as $|e_A| \leq 1 < M/2$). The logarithmic score on dithered cells of width $\Delta < \Delta_c$ and the CRPS rank the two ideal forecasts oppositely, by (a).

By Proposition A.46, neither pair of prices in (a) and (b) is SAI's rate for both laws on the same data. When both laws are computable and, as in (c), the pair of predictions determines q by a computable rule, a theory machine that states both laws and the rule proves every answer, at rate 0. Non-nested information sets are one of the known cases in which rankings of forecasts depend on the loss (for point forecasts, Patton, 2020); (c) is an exact instance in which no forecaster is miscalibrated relative to its own information.

Example A.48 (Local and distance-sensitive scores; misspecification)**.** (a) With the outcome 0, the uniform forecasts on $\{-2, \ldots, 2\}$ and on $\{-20, -10, 0, 10, 20\}$ have the same logarithmic score, $\log 5$ bits, and CRPS 0.4 and 4.0; a forecast with mass 0.21 on 0 and 0.1975 on each of ± 10, ± 20 has a lower logarithmic score (2.2515 bits) than the first and a higher CRPS (3.9006). (b) Let the truth be $\frac{1}{2}N(-1, 0.1^2) + \frac{1}{2}N(1, 0.1^2)$ and the forecasts $N(0, s^2)$. The expected logarithmic score is least at $s = (\mathbb{E}X^2)^{1/2} = 1.004988$, the Kullback–Leibler projection, and the expected CRPS at $s = 1.202984$. The first forecast has expected logarithmic score 2.05427 bits (of the density) against 2.09580 for the second, and expected CRPS 0.60447 against 0.59725: each proper score prefers its own projection, although every strictly proper score ranks the truth first. All numbers are from a direct computation (code in the accompanying repository).

The log-loss posterior over a misspecified family concentrates near the Kullback–Leibler projection (Kleijn and van der Vaart, 2006; cf. Berk, 1966), so asymptotically its predictions are those of the first forecast of (b), whatever the purpose.

Proposition A.49 (Threshold purposes and dominance)**.**

(a) Let $B \supseteq$ RCF *have constants q_t for the times t of a finite domain W, let f be a computable law with rational values on W, $\delta \geq 0$ rational and $k \leq |W|$. Then "for every $t \in W$, $|q_t - f(t)| \leq \delta$" (the honest law T^δ on W) and "for at least k of the $t \in W$, $|q_t - f(t)| \leq \delta$" are sentences, so SAI holds posterior credences in them (*Bel_D*).*

(b) Fix a family of questions, each about one real quantity, and say that law A dominates *law B if for every question A's answer lies in the closed interval between the true value and B's answer. Then A is at least as good as B for every* accuracy-monotone *purpose: per-question losses $\mathsf{L}_Q(a, x)$ that do not increase as the answer a moves towards the true value x from either side, aggregated by any rule that is non-decreasing in each per-question loss (a sum, a maximum, a quantile, a count of failures).*

(c) *Dominance is invariant under every strictly monotone reparametrisation of each quantity separately, but not under a change of the quantities: with truth* $(0,0)$*,* $A=(1,0)$ *dominates* $B=(2,-2)$ *in the quantities* (x,y)*, while in the quantities* $(x,x+y)$ *the predictions are* $(1,1)$ *and* $(2,0)$*, and* B *is exact on the second.*

Proof. (a) Each condition is a finite Boolean combination (for (a)'s second sentence, a disjunction over the k-subsets of W) of order relations between the constants q_t and rationals. (b) Per question $\mathsf{L}_Q(a_A,x)\le\mathsf{L}_Q(a_B,x)$ by monotonicity; then aggregate. (c) A strictly monotone map preserves betweenness; the example is direct. □

A graded purpose, a real-valued loss of the answers or its expectation, is a functional, not a sentence, and SAI has no place for it. Given the quantities, dominance is purpose-neutral, and a dominant law is best for every accuracy-monotone purpose; purposes enter in completing dominance to a ranking and in the choice of the quantities, as in the longitudes and distances of Mars (**??**). That closeness rankings of predictions can reverse under a change of quantities is a classical observation (cf. Miller, 1975). The calibrated pricing can reverse dominance: in Example A.47(b) A' dominates B', yet below Δ'_c the pricing prefers B', so there it is the ranking of no accuracy-monotone purpose.

Rival fits of Kepler's laws. For each of the 39 series of Table 1 we kept the longitude errors of all four fits, 234 pairs of fits in all, and compared the fits by the maximum, RMS and mean absolute error and by the calibrated pricing. The pricing was estimated three ways: by the 40-bin histogram machine SH fitted and scored on disjoint halves of the times, by the same machine with 160 bins on further splits, and by the exact rate $r_{\mathcal{F}}(\Delta)$ of each empirical error law through Lemma A.44, at cell widths from $0.1'$ to $1800'$; a difference is counted only when it is significant by the estimator's rule (for SH, more than two paired standard errors with the same sign on five random half-splits). A computation (code in the accompanying repository) finds the following. The classical criteria disagree with each other: the maximum against the mean absolute error on 204 of the 234 pairs, the maximum against the RMS on 121, the RMS against the mean absolute error on 101. The calibrated pricing agrees with none of them; in the coarse regime it is close to, but not the same as, a ranking by the standard deviation of the errors (with the exact rate at $1800'$ it disagrees with the maximum error on 67 of the 165 decided pairs, with the mean absolute error on 85, with the RMS on 14 and with the standard deviation on 13). Its own ranking of the fits reverses between cell widths for 11 to 16 of the 234 pairs, depending on the estimator and the split; for 13 of the 14 reversals found with the exact rate the coarse-end difference is 0.7% to 6.8% of the rate. Where its significantly best fit is none of the classical winners, the case is a near-tie between the two least-squares fits (0.01% to 0.1% of the rate). Wherever SH and the exact rate both name a significantly best fit, they name the same one.

Definition A.50 (Purpose-SAI)**.** Let $\mathsf{L}:\mathsf{A}\times[-\mathsf{Y},\mathsf{Y}]\to[0,\infty]$ be a loss on a set A of acts, *η-mixable* for an $\eta>0$: for every probability π on A some $\alpha\in\mathsf{A}$ has $\mathsf{L}(\alpha,o)\le-\frac{1}{\eta}\ln\int e^{-\eta\mathsf{L}(\alpha',o)}\pi(d\alpha')$ for every o. The data are answered questions $(\mathcal{Q}_t,a_t)$ (Definition 1.1), each answer a carrying a value $m(a)\in[-\mathsf{Y},\mathsf{Y}]$ (for instance the midpoint of an interval cell), and the outcome at step t is $o_t:=m(a_t)$. After data D the *answer distribution* of a theory machine G on $\mathcal{Q}$ is $P_G(a\mid D,\mathcal{Q}):=\nu_G(D+(a,1))/\sum_{a'\in\mathcal{Q}}\nu_G(D+(a',1))$ when the denominator is positive, and the *act* of G is a Bayes act of L under the induced law of $m(a)$ (for square loss, its mean), or a fixed default act otherwise. *Purpose-SAI* is the aggregating algorithm (Vovk, 1990, 2001) over the theory machines G_i with these acts, learning rate η and prior weights $\mathbf{m}(i)$: at step t it

plays an act α_t that satisfies the mixability inequality for the law π_t that puts mass proportional to $\mathbf{m}(i)e^{-\eta \mathsf{L}_{t-1}(G_i)}$ on the act of G_i, where $\mathsf{L}_t(G)$ is the cumulative loss of G's acts on the first t data.

Proposition A.51 (Regret of purpose-SAI)**.** *If L is η-mixable, then on every sequence of answered questions, for every theory machine G and every n, purpose-SAI has cumulative loss at most $\mathsf{L}_n(G) + K(G)\ln 2/\eta$. Square loss on $[-\mathsf{Y}, \mathsf{Y}]$ is η-mixable for $\eta = 1/(2\mathsf{Y}^2)$, and the regret is then at most $2\mathsf{Y}^2 K(G)\ln 2$.*

Proof. At step t mixability gives $e^{-\eta \mathsf{L}(\alpha_t, o_t)} \geq \sum_i \mathbf{m}(i)e^{-\eta \mathsf{L}_t(G_i)} / \sum_i \mathbf{m}(i)e^{-\eta \mathsf{L}_{t-1}(G_i)}$. The product over $t \leq n$ telescopes, and since $\sum_i \mathbf{m}(i) \leq 1$ the cumulative loss is at most $-\frac{1}{\eta}\ln \sum_i \mathbf{m}(i)e^{-\eta \mathsf{L}_n(G_i)} \leq \mathsf{L}_n(G_i) + \ln(1/\mathbf{m}(i))/\eta$ for every i (the known bound of the aggregating algorithm with prior weights; Vovk, 2001). Take i with $G_i = G$ and $K(i) = K(G)$, so that $\mathbf{m}(i) \geq 2^{-K(G)}$. Square loss on $[0, 1]$ is 2-mixable (cf. Haussler et al., 1998); rescaling $[0, 1]$ to $[-\mathsf{Y}, \mathsf{Y}]$ multiplies the loss by $4\mathsf{Y}^2$ and divides η by the same factor. □

The bound needs an a priori range and grows with its square: for longitudes ($\mathsf{Y} = 10800'$) one bit of $K(G)$ costs about 1.6×10^8 square arcminutes of cumulative square loss, so it is informative only for losses on a range comparable to the errors. Absolute loss is mixable for no η: for outcomes in $\{0, 1\}$ and $\pi = \frac{1}{2}(\delta_0 + \delta_1)$ an act α would need $|\alpha| + |1 - \alpha| \leq -\frac{2}{\eta}\ln\frac{1+e^{-\eta}}{2} < 1$. Like SAI, purpose-SAI is not computable, since its acts need ν_G. After n data its weights are proportional to $\mathbf{m}(i)e^{-\eta \mathsf{L}_n(G_i)}$, the form of the generalised Bayesian posteriors of Bissiri et al. (2016): credences, if anything, about which machine minimises the expected purpose loss, not about which machine is true. For the logarithmic loss with $\eta = \ln 2$ the aggregating algorithm is Bayes. With universal weights over predictors of sequences, the aggregating algorithm underlies predictive complexity (cf. Vovk and Watkins, 1998); only the version over theory machines is new here.

Remark A.52 (A decision layer). Accepting or using an approximate law is an act. Given a posterior predictive of the true answers and a purpose loss on a distribution of questions, laws can be ranked by their posterior-expected purpose loss. This ranks approximate laws by purpose without changing credence, and, if the posterior concentrates on the truth, by their purpose loss against the truth: by maximum error for a tolerance purpose, by RMS error for squared error. For a bounded inducer with a restricted, misspecified pool the posterior predictive is that of the Kullback–Leibler projection (Example A.48(b)), and its acts need not serve the purpose; then the purpose loss must enter learning, as in purpose-SAI or in generalised Bayesian updating. SAI as defined has no act layer, and adding one is not immediate: its predictions are belief functions, so a Bayes act needs a choice, such as the lower expectation or a completion P_D between Bel_D and Pl_D (**??**(c)). For Solomonoff's mixture in sequence prediction, acting Bayes-optimally under a bounded loss has cumulative expected loss close to that of the informed predictor when the true distribution is in the class (Hutter, 2003); for M_A this is open (Question A.53). A user may also prefer a simple approximate law for the cost of computing, communicating or teaching with it; that trade-off is outside both routes.

Question A.53. Can a purpose enter SAI? (i) How should acts be scored under SAI's belief-function predictions (for example by lower expectation), and do the resulting acts serve every bounded purpose loss when the data come from a computable law, as for Solomonoff's mixture? (ii) Which properties of SAI survive in purpose-SAI, for example its equivalence with consistent SOI (**??**), and is there an upper semicomputable universal element: a superloss process over theory machines whose loss is within $O(K(G))$ of every theory machine's loss, as predictive complexity is for sequence prediction?

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